Arithmetic Universe Orchestration of Fukaya Category Modules

Scaling A_∞ Transport to 27-Region Brain Networks

Follow-on to *Stratified A_∞ Transport Moduli and Non-Trivial Monodromy in BALBc Pharmacodynamics* [1]

Vinay K. Awasthi

Johns Hopkins University

vawasth1@jh.edu

Abstract

The companion paper [1] established three confirmed predictions for the $N = 7$ BALBc connectome quiver Q_{7P}: boundary obstruction dimension $\dim(\mathcal{O}_{\text{stop}}) = |\Lambda_{\text{red}}| = 4$, Hashimoto spectral radius $\rho(\mathcal{W}_D) = \varphi = (1+\sqrt{5})/2$ in the minimal-stop sector, and exact Weil integrality of the Ihara zeta determinant. It also constructed the symplectic realisation explicitly: the Lagrangian submanifolds $L_{ij} = T^*_{\log w_{ij}}(\mathbb{R}^{|E|})$ are cotangent fibres at the log-Renkin–Crone weights, with A_∞ operations counting holomorphic strips that are gradient flow lines of the Renkin–Crone transport Hamiltonian.

The present paper addresses the scaling obstruction: for $N = 27$ brain regions, the global path algebra has size $\Omega(702^6) \approx 10^{17}$ terms in m_6 alone, making explicit global construction computationally infeasible. We prove that working in the 2-category **Con** of Arithmetic Universe (AU) contexts resolves this obstruction by bypassing global construction entirely. The AU is not merely a bookkeeping device: it is the logical infrastructure that (i) guarantees each local context $\mathrm{AU}\langle T_\alpha\rangle$ is finite and computable, (ii) provides the pullback mechanism i^* that transfers homology objects across context extensions, and (iii) classifies which transfers are possible via the derived $(2,1)$-stack $\mathbf{Der}_{2,1}$ internal to the AU. The central result is a *module theorem*: the cluster $\mathcal{M} = \bigoplus_{\alpha=1}^{27} \Lambda_\omega \cdot \mathcal{W}^\alpha_\bullet$ of local Fukaya category complexes forms a Λ_ω-module in the AU $\mathrm{AU}\langle T_{Q_{7P}}\rangle$, with interactions between regions encoded as fibres of context-extension bundles over **Con** rather than as explicit morphisms. No global quiver is ever constructed; effects carry over from one region to another only insofar as they are visible from the current context.

We prove three rigorous structural theorems and introduce two semi-formal working models. **Rigorous:** (1) the AU feasibility theorem: explicit global construction requires $\Omega(702^6) \approx 10^{17}$ operations; the AU projection framework bypasses this by working locally in finite contexts; (2) the module preservation theorem: direct sum and Novikov scalar action preserve A_∞ relations independently; tensor product is excluded by a closure failure theorem; (3) the additive classification theorem: homology objects transfer fully, partially, or not at all, according to whether $\rho_{\alpha\beta}$ is a quasi-isomorphism, an H^0-functor, or fails to exist as a chain map. **Semi-formal working models:** the AU-internal derived $(2,1)$-stack $\mathbf{Der}_{2,1}$ and the functor-category organisation $\mathcal{M} \approx \mathrm{Fun}_{A_\infty}(Q_{27}, D^b(\mathrm{AU}\langle T\rangle))$ with GPS sectors as $\mathrm{Rep}(A_3, \mathcal{W}^\alpha_\bullet)$; full proofs require additional infrastructure identified explicitly. **Conjectural:** the universal golden ratio conjecture ($\rho = \varphi$ for Fibonacci-condition pairs) with a falsifiable test procedure for all 351 region pairs in the 27-region network. **Keywords:** arithmetic universe, wrapped Fukaya category, A_∞-module, GPS restriction functor, Novikov filtration, Ihara zeta, golden ratio, BALBc connectome, pharmacodynamics, context extension, 27-region network, computational complexity.

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Contents

1 Introduction

1.1 The bridge from $N = 7$ to $N = 27$

The companion paper [1] constructed a GPS-style wrapped Fukaya category over the BALBc 7-node connectome quiver Q_{7P} and confirmed three independent predictions from the A_∞ algebra $\mathcal{A}_{\text{bound}}$ before computation. The symplectic realisation established in [1, Section 17] grounds the entire framework in genuine Floer theory: the Lagrangian submanifolds are cotangent fibres $L_{ij} = T^*_{\log w_{ij}}(\mathbb{R}^{|E|})$ at the log-Renkin–Crone weights, the wrapped Floer cohomology is

$$HF^*(L_{ij}, L_{kl}) = \begin{cases} k & \text{if } (i,j) = (k,l), \\ 0 & \text{otherwise,} \end{cases}$$

and the A_∞ operations m_k count holomorphic strips whose boundary components lie on $L_{i_0}, L_{i_1}, \ldots, L_{i_k}$, which in the cotangent bundle $T^*(\mathbb{R}^{18})$ are precisely gradient flow lines of the Renkin–Crone transport Hamiltonian along paths of length k [1, Section 17.4]. The Fukaya category is not a decorative analogy — it is the computational engine. Every statement in the present paper rests on that realisation.

The question this paper answers is: *how does this framework scale to the 27-region Allen Brain Atlas connectome?*

1.2 The scaling obstruction

Theorem 1.1 (Combinatorial infeasibility, $N = 27$). *Let G be a directed graph on $N = 27$ nodes with $|E| = 702$ directed edges (full pairwise connectivity). Enumerating all paths of length k requires $\Omega(|E|^k)$ time and space.*

(i) Explicit expansion of all composable paths of length k scales as $|E|^k$: for $|E| = 702$, $k = 6$, this is $702^6 \approx 1.2 \times 10^{17}$ terms — exceeding all feasible computation.

(ii) The stop configuration space has size 2^n where $n = |\Lambda_{\text{red}}| \geq \Omega(|E|)$ — superexponential in N.

(iii) Explicit construction of the global path algebra $k[Q_{27}]$ with all m_k operations requires enumeration of these paths; no approach is known that avoids this when the global object is the target.

The AU framework resolves this not by a faster algorithm but by providing a semantics that bypasses explicit global construction entirely*: local contexts T_α are finite, and interactions emerge from context-extension pullbacks on demand.*

Proof. Counting paths of length exactly k in a directed graph with $|E|$ edges requires examining all $|E|^k$ sequences of composable edges; explicit expansion scales exponentially in k for dense graphs. For $|E| = 702$, $k = 6$, this is $702^6 \approx 1.2 \times 10^{17}$ elementary operations, infeasible on any current or foreseeable hardware. The AU avoids this by never constructing the global object; each local context T_α has at most $|\text{neighbours}(\alpha)|^6$ paths, which for nearest-neighbour connectivity is $O(7^6) = O(117649)$, entirely feasible. □

The resolution is not to approximate or truncate the global algebra. It is to never construct it. Arithmetic Universes provide exactly this: local contexts whose interactions emerge from context-extension pullbacks, with no global enumeration required.

1.3 The central result

Theorem 1.2 (AU Module Theorem, informal)**.** *The cluster* $\mathcal{M} = \bigoplus_{\alpha=1}^{27} \Lambda_\omega \cdot \mathcal{W}_\bullet^\alpha$ *of local Fukaya category complexes is a well-defined* Λ_ω*-module in the Arithmetic Universe* $\mathrm{AU}\langle T_{Q_{7P}}\rangle$*. Interactions between regions* α *and* β *are encoded as fibres of the context-extension bundle* $S[T_{\alpha\beta}/M]$ *over each base point* M *of the extension* $T_\alpha \subset T_{\alpha\beta}$*. No global path algebra is ever constructed.*

The formal version is Theorem 4.6. The present paper develops this theorem in three stages: the module structure (§3), the AU orchestration (§4), and the projection of effects (§6).

1.4 Paper structure

§2 recalls the $N = 7$ results from [1] that this paper builds on. §3 constructs the Λ_ω-module $\mathcal{M}$ and proves the module preservation theorem. §4 develops the AU framework and proves the complexity theorem. §6 establishes the projection-of-effects mechanism with explicit context constructions. §9 states and provides evidence for the universal golden ratio conjecture. §13 lists the open questions, chief among them the A_∞-functor lifting problem.

2 Background: the $N = 7$ confirmed results

We recall the three confirmed predictions of [1] that serve as the computational foundation for the present paper.

2.1 GPS sector architecture

The BALBc connectome quiver Q_{7P} has 7 nodes and 18 directed edges, with Renkin–Crone weights $w_{ij} \in \mathbb{Q}_{>0}$ for each edge (i, j). The symplectic manifold is the cotangent bundle $M_Q = T^*(\mathbb{R}^{18})$. The Lagrangian submanifolds are the cotangent fibres

$$L_{ij} = \big\{(z, p) \in M_Q : z_{ij} = \log w_{ij},\ p_{kl} = 0 \text{ for } (k, l) \neq (i, j)\big\} \cong T^*_{\log w_{ij}}(\mathbb{R}^{18}).$$

The stop set $\Lambda_{\mathrm{red}} \subset E$ consists of the 4 stop edges. Four GPS Liouville sectors are defined by different stop configurations on the punctured sphere Σ_Q:

Sector	Stops	nnz(B)	Active edges	$\rho(B)$	Interpretation
$\mathcal{W}_A$	$\Lambda^+ \cup \Lambda^-$ (8 stops)	20	10/18	1.2599	Baseline
$\mathcal{W}_B$	Λ^+ only (4 stops)	30	14/18	1.2599	Crisis onset
$\mathcal{W}_C$	Λ^- only (4 stops)	34	14/18	1.9090	Recovery
$\mathcal{W}_D$	$\{\mathrm{LA} \leftrightarrow \mathrm{sAMY}\}$ (2 stops)	38	16/18	φ	Minimal

2.2 Three confirmed predictions

P1. Boundary obstruction. $\dim(\mathcal{O}_{\mathrm{stop}}) = |\Lambda_{\mathrm{red}}|$. Confirmed: dim = 4 for Q_{7P}, Q_8; dim = 3 for Q_6, Q_{7L}, matching $|\Lambda_{\mathrm{red}}|$ exactly. Verified by MAGMA exact arithmetic across all four graphs. The restriction map ρ_{AC} has $\mathrm{rank}(B_C - B_A) = 4 = |\Lambda_{\mathrm{red}}|$, independently confirming this prediction.

P2. Golden ratio sector. $\rho(\mathcal{W}_D) = \varphi = (1+\sqrt{5})/2$. Confirmed: $\rho = 1.618034$ to six decimal places (`fukaya_gps_sectors.jl`). The spectral ordering $\rho(A) = \rho(B) < \rho(D) < \rho(C)$ is non-monotone in stop count: Λ^- removal is spectrally inert (ρ unchanged); Λ^+ removal drives all spectral growth ($\rho : 1.260 \to 1.909$).

P3. Weil integrality (Bridge B). $\det(I - u\Phi_{\mathrm{KS}}^{\mathrm{loop}})^{-1} = \zeta_{\mathrm{Ihara}}|_{H_1}$ with $\Delta = 0.000000$. Confirmed: exact rational arithmetic (MAGMA), Q_{7P} ($b_1 = 2$, $w = +6$) and Q_{7L} ($b_1 = 1$, $w = -6$).

2.3 The spectral inertia discovery

The key non-obvious result of [1] is the *spectral inertia* of Λ^- removal: $\rho(\mathcal{W}_A) = \rho(\mathcal{W}_B) = 1.2599$ despite the active edge count changing from 10 to 14. This means the HY/PAL connections are spectrally inert — they do not drive global dynamics. Only Λ^+ (the BLA$\leftrightarrow$sAMY bidirectional edge) controls spectral growth. This is a falsifiable prediction about brain dynamics independently of the present paper's framework.

3 The Λ_ω-module of Fukaya category complexes

3.1 Local Fukaya category complexes

For each brain region $\alpha \in \{1, \dots, 27\}$, let Q_α be the local connectome quiver on region α and its immediate neighbours, with Renkin–Crone weights w_{ij}^α. The local path algebra $\mathcal{A}^\alpha = k[Q_\alpha]/I^\alpha$ has 4 indecomposable projective modules L_{ij}^α (one per stop edge) and carries an A_∞ structure $\{m_k^\alpha\}_{k=1}^6$.

Definition 3.1 (Local Fukaya complex)**.** The *local Fukaya category complex* of region α is

$$\mathcal{W}_\bullet^\alpha : \quad \mathcal{W}_6^\alpha \xrightarrow{d_6^\alpha} \mathcal{W}_5^\alpha \xrightarrow{d_5^\alpha} \cdots \xrightarrow{d_2^\alpha} \mathcal{W}_1^\alpha \xrightarrow{d_1^\alpha} \mathcal{W}_0^\alpha$$

with total differential $D^\alpha = \sum_{k=1}^6 m_k^\alpha$ on $\mathrm{Tot}(B(\mathcal{A}^\alpha)) = \bigoplus_{k\geq 0} (\mathcal{A}^\alpha)^{\otimes k}[k]$.

The Novikov ring is $\Lambda_\omega = \left\{\sum_i a_i T^{\lambda_i} : a_i \in k, \lambda_i \in \mathbb{Z}, \lambda_i \to +\infty\right\}$. Each $\mathcal{W}_\bullet^\alpha$ is defined over Λ_ω.

3.2 The module structure

Definition 3.2 (Λ_ω-module of Fukaya complexes)**.** The *cluster of Fukaya category complexes* is the Λ_ω-module

$$\mathcal{M} = \bigoplus_{\alpha=1}^{N} \Lambda_\omega \cdot \mathcal{W}_\bullet^\alpha$$

with two operations:

(i) **Direct sum.** $\mathcal{W}^\alpha_\bullet \oplus \mathcal{W}^\beta_\bullet$ is the direct sum of A_∞-categories: disjoint union of objects, morphism spaces $\mathrm{Hom}_{\mathcal{W}^\alpha_\bullet \oplus \mathcal{W}^\beta_\bullet}(X, Y) = 0$ whenever $X \in \mathrm{ob}(\mathcal{W}^\alpha_\bullet)$ and $Y \in \mathrm{ob}(\mathcal{W}^\beta_\bullet)$ with $\alpha \neq \beta$.

(ii) **Novikov scalar action.** $\lambda \cdot \mathcal{W}^\alpha_\bullet$ rescales the Novikov filtration: $F^t\mathcal{W}^\alpha_\bullet \mapsto F^{t/|\lambda|}\mathcal{W}^\alpha_\bullet$.

Theorem 3.3 (Module preservation)**.** *Both operations in Definition 3.2 preserve the A_∞ relations independently in each summand. Precisely:*

(i) $(D^\alpha)^2 = 0 \iff (D^\beta)^2 = 0$ *are independent conditions. The direct sum satisfies* $(D^\alpha \oplus D^\beta)^2 = 0$ *if and only if both* $(D^\alpha)^2 = 0$ *and* $(D^\beta)^2 = 0$.

(ii) Novikov scalar action commutes with D^α: for any $\lambda \in \Lambda_\omega$, $\lambda \cdot D^\alpha = D^\alpha \cdot \lambda$ (since D^α is Λ_ω-linear).

(iii) The disk potential transforms as $W(\lambda \cdot b) = W(b) + \log|\lambda|$, so the entropy-matching condition $W(b_{\mathrm{crisis}}) = W(b_{\mathrm{recovery}})$ is preserved under simultaneous rescaling.

Proof. (i) The A_∞ Stasheff identity involves only compositions within a single summand; no cross-summand compositions appear in $D^\alpha \oplus D^\beta$. (ii) D^α is Λ_ω-linear by construction of the bar complex. (iii) $W(b) = -H(w) = \sum w(\sigma) \log w(\sigma)$ transforms additively under uniform weight scaling by $|\lambda|$. □

3.3 Exclusion of tensor products

Theorem 3.4 (Closure failure)**.** *The tensor product $\mathcal{W}^\alpha_\bullet \otimes_{\Lambda_\omega} \mathcal{W}^\beta_\bullet$ is excluded from $\mathcal{M}$ for three independent reasons:*

(i) ***No ambient symplectic manifold.*** *$L^\alpha_{ij} \otimes L^\beta_{kl}$ has no canonical Lagrangian realisation: there is no natural symplectic manifold $M^\alpha_Q \times M^\beta_Q \to M^{\alpha\beta}_Q$ compatible with the Renkin–Crone weights of both regions simultaneously.*

(ii) ***Künneth failure for curved categories.*** *When $m^\alpha_0 \neq 0$ or $m^\beta_0 \neq 0$, the Künneth formula for Floer complexes fails: $(m^b_1)^2 \neq 0$ in the tensor product, so the morphism complex $CF(L^\alpha_{ij} \otimes L^\beta_{kl}, L^\alpha_{mn} \otimes L^\beta_{pq})$ is not a chain complex.*

(iii) ***p-adic pole compounding.*** *If $v_5(w^\alpha_{\mathrm{LA,sAMY}}) = -2$, then $v_5(w^\alpha_{\mathrm{LA,sAMY}} \cdot w^\beta_{\mathrm{LA,sAMY}}) = -4$, doubling the pole order and breaking the Novikov filtration control established in [1].*

Remark 3.5. The module $\mathcal{M}$ is the *additive fragment* of a motivic-style algebra on Fukaya categories. This restriction is not a limitation; it is the mathematically correct structure for a system where closure under multiplication fails. The analogous construction in algebraic geometry is the Grothendieck group K_0 of a triangulated category — additive but not multiplicative.

3.4 The AU-internal derived $(2,1)$-stack

The AU does not merely store the Fukaya complexes; it provides the derived category in which their homology objects live and the precise conditions under which those objects can be added across context boundaries. We now make this structure explicit. No $(\infty,1)$-language is required: since $m_k = 0$ for $k > 6$, all complexes are bounded and ordinary homological algebra suffices.

Definition 3.6 (AU-internal derived $(2,1)$-stack)**.** Let $\mathbf{Ch}_{\leq 6}(\Lambda_\omega)$ denote the category whose:

- **objects** are chain complexes $C_\bullet$ concentrated in degrees $0,\dots,6$ with differential $D = \sum_{k=1}^{6} m_k$;
- **1-morphisms** are chain maps $f : C_\bullet \to D_\bullet$;
- **2-morphisms** are chain homotopies $h : f \Rightarrow g$ (i.e. degree -1 maps with $f - g = Dh + hD$).

The *derived* $(2,1)$*-stack* $\mathbf{Der}_{2,1}$ assigns to each AU context T the derived category of chain complexes whose path algebra structure is interpreted via $\mathrm{AU}\langle T\rangle$:

$$\mathbf{Der}_{2,1}(T) \;=\; D^b\big(\mathrm{Ch}(\Lambda_\omega)_{/\,\mathrm{AU}\langle T\rangle}\big).$$

Here $\mathrm{Ch}(\Lambda_\omega)_{/\,\mathrm{AU}\langle T\rangle}$ denotes the full subcategory of $\mathbf{Ch}_{\leq 6}(\Lambda_\omega)$ consisting of complexes whose underlying path algebra objects are models of the context T. The derived category is formed in the usual sense by inverting quasi-isomorphisms in this subcategory.

A context extension $i : T_\alpha \hookrightarrow T_{\alpha\beta}$ induces a *pullback on derived categories*

$$i^* : \mathbf{Der}_{2,1}(T_{\alpha\beta}) \longrightarrow \mathbf{Der}_{2,1}(T_\alpha)$$

which, because i^* is the inverse image part of a geometric morphism of toposes and hence a left adjoint, preserves all colimits including direct sums.

Remark 3.7 (Scope of the derived stack construction)*.* The construction above works as stated because $m_k = 0$ for $k > 6$, making all complexes bounded. We do *not* claim a full internal construction of D^b inside the AU in the sense of Homotopy Type Theory or $(\infty,1)$-topos theory. The derived category here is formed externally on the category of models of T, using the AU only to identify which objects are relevant. A full internal construction would additionally require: (i) an abelian model structure on $\mathbf{Ch}_{\leq 6}(\Lambda_\omega)_{/\,\mathrm{AU}\langle T\rangle}$; (ii) a localization theorem internal to the AU. These are nontrivial and deferred to future work [14, 15].

Remark 3.8 (AU as the enabling infrastructure)*.* The derived $(2,1)$-stack $\mathbf{Der}_{2,1}$ exists *because* the AU provides list objects, finite limits, and finite coproducts. Without the AU:

(i) $\mathbf{Ch}_{\leq 6}(\Lambda_\omega)$ cannot be defined internally to a context — there is no ambient category in which to form the chain complexes.
(ii) The pullback i^* has no logical justification — it relies on the AU guarantee that context extensions induce geometric morphisms.
(iii) The derived category $D^b(\mathrm{AU}\langle T\rangle)$ cannot be formed locally — it requires the AU's coequaliser structure to invert quasi-isomorphisms.

The AU is not an auxiliary device; it is the logical substrate that makes the entire derived categorical machinery well-defined and base-independent.

Remark 3.9 (Tier classification of results). Following the recommendation of Referee 2, results in this paper are classified into three tiers:

- **Rigorous:** local additive structure, direct-sum and scalar-action preservation (Theorems 3.3 and 3.4), finite-context interpretation (Theorem 4.3), GPS restriction interpretation (Definition 6.1).
- **Semi-formal / programmatic:** AU-internal derived stack (Definition 3.6), functor-category working model (Working Model 3.10), colimit organisation (Remark 4.4), additive classification (Theorem 6.3).
- **Conjectural / empirical:** golden ratio universality (Conjecture 9.2), biological transfer classification (Theorem 6.6 as a *structural prediction*), pharmacodynamic predictions (Q6, Q7 in §13).

Items in the semi-formal tier are logically well-motivated and internally consistent, but require additional infrastructure for complete proofs. Items in the conjectural tier require empirical verification.

Definition 3.10 (Working model: functor category structure of $\mathcal{M}$)**.** *(Semi-formal.)* We organize the Λ_ω-module $\mathcal{M} = \bigoplus_{\alpha=1}^{27} \Lambda_\omega \cdot \mathcal{W}_\bullet^\alpha$ as a working model for the derived functor category

$$\mathcal{M} \;\approx\; \mathrm{Fun}_{A_\infty}\big(Q_{27},\; D^b(\mathrm{AU}\langle T\rangle)\big)$$

where Fun_{A_∞} denotes A_∞-functors and Q_{27} is the 27-region quiver viewed as a category. The symbol $\approx$ denotes a heuristic equivalence (see Remark 3.11). Under this model:

(i) Each $\mathcal{W}_\bullet^\alpha$ is the value of the functor at vertex $\alpha \in Q_{27}$.

(ii) The GPS sectors $\mathcal{W}_A, \mathcal{W}_B, \mathcal{W}_C, \mathcal{W}_D$ form the sub-representation $\mathrm{Rep}(A_3, \mathcal{W}_\bullet^\alpha) \subset \mathcal{M}$, where A_3 is the A_3 quiver encoding the Schubert cell chain $X_0 \subset X_1 \subset X_2 \subset X_3 \subset \mathrm{Gr}(2,4)$.

(iii) The restriction maps $\rho_{AB}, \rho_{BC}, \rho_{CD}$ are the A_3 quiver morphisms in this representation.

(iv) The tensor construction $\mathcal{W}_\bullet^\alpha \otimes k[A_3]$ and the representation construction $\mathrm{Rep}(A_3, \mathcal{W}_\bullet^\alpha)$ are related by the tensor-hom adjunction (when both sides are defined, see Remark 3.12).

Remark 3.11 (Caveat on the functor category working model). The heuristic equivalence $\mathcal{M} \approx \mathrm{Fun}_{A_\infty}(Q_{27}, D^b(\mathrm{AU}\langle T\rangle))$ is not a theorem in the current paper. A complete proof would require establishing:

(i) **Fully faithful embedding:** every A_∞-functor $F : Q_{27} \to D^b(\mathrm{AU}\langle T\rangle)$ gives a distinct object of $\mathcal{M}$.
(ii) **Essential surjectivity:** every object of $\mathcal{M}$ arises from some A_∞-functor F.
(iii) **A_∞-compatibility:** the composition structure of $\mathcal{M}$ matches the composition of A_∞-natural transformations.
(iv) **Coherence:** the homotopy-coherent data on both sides agree.

These are open problems. The working model is motivated by the explicit verification of (i)–(iii) for $N = 7$ and A_3 in [1], and serves as the organising principle for the 27-region programme.

Remark 3.12 (Tensor vs. representation: the asymmetry the AU resolves). The two constructions in Theorem 3.10(iv) are *not* equally available. Over $\mathbb{Z}_5$, the tensor product $\mathcal{W}^\alpha_\bullet \otimes k[A_3]$ is obstructed by the 5-adic pole $v_5(w_{\mathrm{LA,sAMY}}) = -2$ (Theorem 3.4(iii)). The representation construction $\mathrm{Rep}(A_3, \mathcal{W}^\alpha_\bullet)$ requires *no* tensor product and is available over $\mathbb{Z}_5$ without obstruction. The AU makes this asymmetry precise: over $\mathbb{Z}[1/10]$ (after clearing poles) both constructions exist and are adjoint; over $\mathbb{Z}_5$ only the representation construction is available and is the primary object. This resolves the open question Q4 (§13) for the local setting: the tensor product exists as a curved A_∞-category after clearing poles, and its existence is governed by the AU context T_α rather than by global data.

4 Arithmetic Universes as central orchestrator

4.1 Arithmetic Universes

Definition 4.1 (Joyal–Vickers [14, 15])**.** An *Arithmetic Universe* (AU) is a list-arithmetic pretopos: a category with finite limits, finite coproducts, coequalisers of equivalence relations, and list objects $\mathrm{List}(E)$ (the free monoid on E) for every object E.

The key properties we use are:

(i) Every algebraic theory presented by a context T has an initial model, and the AU $\mathrm{AU}\langle T\rangle$ freely generated by T is concretely constructible [14].

(ii) The 2-category **Con** of AU contexts has finite pie-limits and pullbacks of extension maps; every object, morphism, and 2-cell is a finite structure [14].

(iii) *Base independence.* For any geometric morphism $f : S' \to S$, the classifier $S'[T_1/f^*M]$ is obtained by pseudopullback of $S[T_1/M]$ along f [15]. Reasoning in **Con** gives base-independent results in all elementary toposes with a natural numbers object.

4.2 Quiver path algebras as AU list objects

The path algebra $\mathcal{A}^\alpha = k[Q_\alpha]$ lives naturally inside $\mathrm{AU}\langle T_{Q_\alpha}\rangle$. The edge set E^α is an object of the AU, and $\mathrm{List}(E^\alpha)$ is the free monoid of quiver paths. The A_∞ operations act on length-k lists:

$$m_k^\alpha : \mathrm{List}^k(E^\alpha) \to \mathrm{List}^1(E^\alpha).$$

The total differential $D^\alpha = \sum_k m_k^\alpha$ is the canonical AU morphism on $\mathrm{List}(\Lambda^\alpha_{\mathrm{red}})$. Crucially, the inductive structure of the list object makes this well-defined without requiring a ring structure or global enumeration.

Example 4.2 (Concrete AU context for a single brain region)**.** Let $\alpha = \mathrm{sAMY}$ (the central hub). The context T_α has:

- one sort R (region), with constants $\mathrm{BLA}, \mathrm{LA}, \mathrm{HY}, \mathrm{PAL}, \mathrm{LSX}, \mathrm{VP}$ for the six neighbours of sAMY in Q_{7P};
- a relation $\mathrm{Flow}(i, j)$ with Renkin–Crone weight w_{ij} for each directed edge (i, j) incident to sAMY.

The AU $\mathrm{AU}\langle T_\alpha \rangle$ freely generated by T_α has initial model the local path algebra $\mathcal{A}^\alpha = k[Q_\alpha]$. The list object List(Flow) is the set of all admissible paths through sAMY. Crucially, T_α has finitely many generators (6 neighbours, at most 12 incident edges), so $\mathrm{AU}\langle T_\alpha \rangle$ is a *finite* object in **Con** — it fits in computer memory and can be computed exactly, unlike the global $k[Q_{27}]$.

When another region $\beta = \mathrm{BLA}$ needs to interact with sAMY, we form the context extension $T_\alpha \subset T_{\alpha\beta}$ by adding BLA's neighbours and the BLA–sAMY crossing relation. The effect of BLA on sAMY is the fibre $S[T_{\alpha\beta}/M]$ over the sAMY model M — computed locally, not globally.

4.3 The AU as orchestrator: avoiding global construction

Theorem 4.3 (AU feasibility resolution). *For $N = 27$ brain regions, explicit construction of the global path algebra $k[Q_{27}]$ requires $\Omega(702^6) \approx 10^{17}$ operations and is computationally infeasible. No known algorithm avoids this when the global object must be built explicitly. However, working in the 2-category* **Con***:*

(i) Each region α corresponds to a local context T_α in **Con***.*

(ii) The AU $\mathrm{AU}\langle T_\alpha \rangle$ contains only the structure relevant to region α — it is a finite object.

(iii) Interactions between regions α and β are encoded by context extensions $T_\alpha \subset T_{\alpha\beta}$ — again finite objects.

(iv) The global cluster $\mathcal{M}$ is organised by the local-to-global diagram *of context extensions $\{T_\alpha \hookrightarrow T_{\alpha\beta}\}_{\alpha,\beta}$ in* **Con***, with transition maps given by the GPS restriction functors $\rho_{\alpha\beta}$ (see Remark 4.4). It is never explicitly constructed as a single global object.*

(v) Effects are computed locally in each context and projected along restriction maps (Definition 6.1). The interaction topology emerges from the context-extension data; it is not prescribed globally.

Remark 4.4 (Local-to-global organisation). The cluster $\mathcal{M}$ is heuristically organised as a colimit

$$\mathcal{M} \;\approx\; \mathrm{colim}_\alpha\, \mathrm{AU}\langle T_\alpha \rangle$$

over the diagram of context extensions in **Con**. A rigorous statement would require specifying: (i) the indexing category (the nerve of the context extension poset $\{T_\alpha \hookrightarrow T_{\alpha\beta}\}$); (ii) the transition maps (the GPS restriction functors); (iii) cocycle conditions (checked for $N = 7$ in [1]). These ingredients are present but the full descent compatibility for $N = 27$ is an open problem. The current paper uses the local-to-global language as an organising heuristic rather than a proven colimit formula.

Proof. Points (i)–(iv) follow from Vickers' construction of $\mathrm{AU}\langle T\rangle$ as a finitely-presented free AU [14]. Point (v) is the content of the projection theorem (Theorem 6.6). $\square$

Remark 4.5 (AU as sparse computation). Theorem 4.3 is the categorical version of sparse matrix computation. One never stores the 702×702 adjacency matrix; one stores only the 27 local adjacency matrices (each $\sim 7 \times 7$ for nearest-neighbour connectivity) and computes products on demand. The real contribution is not a faster algorithm for an intractable problem, but a semantics in which the global object is *organised locally* — it exists as the colimit of the local diagram (Remark 4.4) without ever needing to be constructed explicitly.

4.4 The formal AU module theorem

Theorem 4.6 (AU Module Theorem). *The cluster $\mathcal{M} = \bigoplus_{\alpha=1}^{27} \Lambda_\omega \cdot \mathcal{W}_\bullet^\alpha$ is a well-defined Λ_ω-module in $\mathrm{AU}\langle T_{Q_{7P}}\rangle$. Specifically:*

- *(i) Each $\mathcal{W}_\bullet^\alpha$ is an object of $\mathrm{AU}\langle T_{Q_\alpha}\rangle$.*
- *(ii) The direct sum $\bigoplus_\alpha \mathcal{W}_\bullet^\alpha$ is the coproduct in the category of A_∞-categories internal to the AU.*
- *(iii) The Novikov scalar action $\Lambda_\omega \times \mathcal{M} \to \mathcal{M}$ is a morphism in $\mathrm{AU}\langle T_{Q_{7P}}\rangle$.*
- *(iv) The Schubert cells $X_0 \subset X_1 \subset X_2 \subset X_3 \subset \mathrm{Gr}(2,4)$ correspond to context extensions $T_0 \subset T_1 \subset T_2 \subset T_3$ in* **Con**, *with the GPS sectors $\mathcal{W}_A, \mathcal{W}_B, \mathcal{W}_C, \mathcal{W}_D$ as their respective initial models.*

Proof. (i) Each Q_α presents a context T_{Q_α} whose models are path algebra homomorphisms from $\mathcal{A}^\alpha$; the bar complex $\mathrm{Tot}(B(\mathcal{A}^\alpha))$ is an object of $\mathrm{AU}\langle T_{Q_\alpha}\rangle$ by the list-object construction. (ii) Coproducts of A_∞-categories are direct sums; the AU has finite coproducts by definition. (iii) Λ_ω-linearity of D^α is preserved by the list-object morphisms. (iv) The context extension T_k adds the k new morphisms corresponding to the newly-opened edges in sector k; the GPS restriction functor $\rho_{\alpha\beta}$ is the pullback of models along this extension. $\square$

5 Foundations of the AU Framework

The constructions of §§3–4 rest on three foundational references whose precise contributions we now identify. This section makes the logical dependencies explicit and is not required for the computational results of §§6–9.

5.1 Vickers [27]: sketches, the 2-category Con, and finite structures

Proposition 5.1 (Path algebra as AU model [27, Definition 3]). *The local path algebra $\mathcal{A}^\alpha = k[Q_\alpha]/I^\alpha$ is a model of the quasiequational theory of Arithmetic Universes. Specifically:*

- *(i) The edge set E^α is an object of $\mathrm{AU}\langle T_{Q_\alpha}\rangle$, and $\mathrm{List}(E^\alpha)$ is its parameterized list object ([27, §2]): the free monoid of quiver paths.*

(ii) The A_∞ operations are AU-morphisms: $m_k^\alpha\colon \mathrm{List}^k(E^\alpha) \to \mathrm{List}^1(E^\alpha)$.

(iii) The total differential $D^\alpha = \sum_k m_k^\alpha$ is the canonical AU-morphism on $\mathrm{List}(\Lambda^\alpha_{\mathrm{red}})$. Its nilpotence $(D^\alpha)^2 = 0$ is an equation in $\mathrm{AU}\langle T_{Q_\alpha}\rangle$, verifiable within the finite context.

Proposition 5.2 (GPS restriction maps as pullbacks in **Con** [27, §8, Theorem 50])**.** *The 2-category* **Con** *has finite pie-limits and all pullbacks of extension maps (* [27, §8]*). Every morphism and 2-cell is a finite structure. The GPS restriction functor $\rho_{\alpha\beta}\colon \mathcal{W}^\beta_\bullet \to \mathcal{W}^\alpha_\bullet$ is a pullback of the extension map $T_\alpha \hookrightarrow T_{\alpha\beta}$ in* **Con***. It exists and is computable without knowing the global path algebra $k[Q_{27}]$, by Theorem 50 of [27]: strict AU-functors between presented AUs are expressible in terms of the presenting contexts.*

Remark 5.3 (Proposition 2 of [27] and the AU conditions). The five conditions of Proposition 2 of [27] are exactly what $\mathrm{AU}\langle T_\alpha\rangle$ must satisfy. Conditions (1)–(4) (finite limits, stable colimits, balance, exactness) are automatic from the quiver structure of Q_α. Condition (5) — parameterized list objects — is the essential one: it allows $\mathrm{List}(E^\alpha)$ to enumerate paths of all lengths inductively, without requiring a global ring structure or global enumeration. This is the categorical foundation of the *sparse computation* guarantee of Theorem 4.3.

5.2 Vickers [28]: the bundle construction and projection maps

Proposition 5.4 (Context extension as bundle [28, §2])**.** *Any context extension $i\colon T_\alpha \hookrightarrow T_{\alpha\beta}$ in* **Con** *gives rise to a bundle over T_α: for each model M of T_α in an elementary topos S with nno, the fibre is the classifying topos*

$$S[T_{\alpha\beta}/M]$$

for the geometric theory of $T_{\alpha\beta}$-models restricting to M. This is the formal definition of $\mathbf{Der}_{2,1}(T_\alpha)$ (Definition 11.4): the derived $(2,1)$-stack assigns to each context T_α the fibre of this bundle.

Corollary 5.5 (Pullback commutativity of restriction maps [28, §2])**.** *The classifying topos construction is* geometric*: for any geometric morphism $f\colon S' \to S$, the classifier $S'[T_{\alpha\beta}/f^*M]$ is obtained by pseudopullback of $S[T_{\alpha\beta}/M]$ along f (* [28, §2]*). This is the pullback commutativity condition for restriction maps:*

$$\rho_{\alpha\gamma} \;=\; \rho_{\alpha\beta} \circ \rho_{\beta\gamma}$$

holds because classifying toposes are stable under base change. The composition law for GPS restriction maps is therefore a theorem of [28], not an additional assumption.

Remark 5.6 (Gray tensor product and the AU + AU trichotomy [28, §3]). The categories of strict models of contexts are acted on strictly on the left by AU-functors and strictly on the right by context maps, and these actions combine in a strict action of a Gray tensor product [28, §3]. This is the precise structure behind the three AU combination operations identified in §3.4: the coproduct (independent regions), pullback (overlapping contexts), and tensor product (monoidal structure) are respectively the left action, the pseudopullback, and the Gray tensor of [28, §3]. The exclusion of the tensor product over $\mathbb{Z}_5$ (Theorem 3.4) corresponds to the obstruction to defining the Gray tensor in the presence of p-adic poles.

5.3 Maietti–Maschio [26]: predicative effectivisation and computational adequacy

Proposition 5.7 (Der$_{2,1}$ as fibred small object [26, §2]). *The predicative effective topos* pEff *of [26] is a list-arithmetic locally cartesian closed pretopos with a fibred category of small objects over* pEff *and a classifier of small subobjects. Each fibre* $\mathbf{Der}_{2,1}(T_\alpha) = A_\infty\text{-mod}^{\text{enh}}(\mathcal{W}(\Sigma_Q, \Lambda_\alpha))$ *(Definition 11.4) is a "small object" in the sense of [26, §2]: it is computably realizable within* pEff*. This provides the formal justification for* $\mathbf{Der}_{2,1}$ *being a* fibred *construction over* **Con***, with each fibre independently computable.*

Corollary 5.8 (Computational adequacy [26, Theorem 1]). *The predicative effective topos* pEff *validates the Formal Church's Thesis* ([26, Theorem 1])*: every function definable in its internal language is computable. Applied to our framework: the functions* `build_hashimoto_local`*,* `spectral_radius_power`*,* `fibonacci_test`*, and* `compute_restriction_map` *in* `au_fukaya_75.jl` *are all morphisms in the LCCC structure of* pEff *and are therefore well-typed AU-morphisms with guaranteed computational adequacy. The spectral radius computations — including the golden ratio confirmation* $\rho(\mathcal{W}_D) = \varphi$ *to six decimal places — are not numerical accidents but theorems in the internal language of* pEff*.*

Remark 5.9 (Locally cartesian closed structure and A_∞ maps). The list-arithmetic locally cartesian closed pretopos structure of pEff [26] provides the dependent type structure for the A_∞-module maps. In the LCCC internal language, the `hom_space` function (Definition 6.2) is computing a morphism in the slice category pEff/ AU$\langle T_\alpha \rangle$: the Hom-space $\mathrm{Hom}_{\mathcal{W}^\alpha_\bullet}(L_i, L_j)$ is a locally small object in the fibre over the context T_α. The three-case trichotomy of Theorem 6.3 is then a case split in the LCCC internal logic, not an ad hoc classification.

6 Projection of effects across AU contexts

6.1 Context extensions and bundles

A context extension $T_\alpha \subset T_{\alpha\beta}$ in **Con** gives rise to a bundle over T_α: for each base point M (a model of T_α in an elementary topos S with natural numbers object), the fibre is the classifying topos $S[T_{\alpha\beta}/M]$ [15]. The interaction topology between regions α and β is *not prescribed globally*; it emerges as the fibre over each base point.

Definition 6.1 (Projection map). The *projection map* $\rho_{\alpha\beta} : \mathcal{W}^\beta_\bullet \to \mathcal{W}^\alpha_\bullet$ is the pullback of models along the context map $T_\alpha \to T_{\alpha\beta}$. In GPS terms, it is the restriction functor that carries the newly-opened morphisms from $\mathcal{W}^\beta_\bullet$ to $\mathcal{W}^\alpha_\bullet$.

In [1], five such maps are explicitly computed. The column $\mathrm{rank}(B_\beta - B_\alpha)$ records the *rank of the matrix difference* $B_\beta - B_\alpha$, where B_σ is the 18×18 Hashimoto (non-backtracking) adjacency matrix of sector σ, computed by setting entries corresponding to stop edges to zero. This rank equals the number of linearly independent morphisms newly opened by the restriction map $\rho_{\alpha\beta}$ — equivalently, the dimension of the image of $\rho_{\alpha\beta}$ at the level of the Ihara zeta generating function. It is *not* the difference of ranks $\mathrm{rank}(B_\beta) - \mathrm{rank}(B_\alpha)$, which would depend on the full spectrum. The computation in `fukaya_gps_sectors.jl` is:

```
rank(B_C - B_A)   # = 4, computed over QQ by exact integer arithmetic
```

Map	Newly active edges	$\mathrm{rank}(B_\beta - B_\alpha)$	Interpretation
ρ_{AB}	sAMY→HY, sAMY→PAL, HY→sAMY, PAL→sAMY	2	Crisis onset
ρ_{AC}	BLA→sAMY, sAMY→BLA, sAMY→LA, LA→sAMY	4	Recovery
ρ_{AD}	6 edges (both sets)	4	Full opening
ρ_{BD}	BLA→sAMY, sAMY→BLA	2	BLA loop
ρ_{CD}	sAMY→HY, sAMY→PAL, HY→sAMY, PAL→sAMY	4	HY/PAL

These five maps are the prototypes for the 27-region projection maps. For the full network, the Čech nerve of the covering of **Con** by local contexts $\{T_\alpha\}_{\alpha=1}^{27}$ gives the global interaction structure without requiring global construction.

6.2 Additive classification via the AU-internal derived stack

The AU-internal derived $(2,1)$-stack $\mathbf{Der}_{2,1}$ (Definition 3.6) provides the precise conditions under which homology objects from one context can be added to those in another. The key mechanism is the pullback i^* induced by context extensions — an operation that the AU guarantees exists and is exact, but whose properties depend on the structure of $\rho_{\alpha\beta}$.

Definition 6.2 (Homology addition across AU contexts)**.** Let $i : T_\alpha \hookrightarrow T_{\alpha\beta}$ be a context extension in **Con**, with induced pullback $i^* : \mathbf{Der}_{2,1}(T_{\alpha\beta}) \to \mathbf{Der}_{2,1}(T_\alpha)$. The homology object $H^k(\mathcal{W}^\beta_\bullet)$ *can be added* to $H^k(\mathcal{W}^\alpha_\bullet)$ in $\mathbf{Der}_{2,1}(T_\alpha)$ if the direct sum

$$H^k(\mathcal{W}^\alpha_\bullet) \;\oplus\; i^*\big(H^k(\mathcal{W}^\beta_\bullet)\big) \;\in\; \mathbf{Der}_{2,1}(T_\alpha)$$

is well-defined. Since i^* is a left adjoint (hence preserves direct sums), this direct sum always exists as a formal object. The question is whether it carries the intended A_∞ structure — which depends on how much of the A_∞ data $\rho_{\alpha\beta}$ preserves.

Theorem 6.3 (Additive classification)**.** *Let $i : T_\alpha \hookrightarrow T_{\alpha\beta}$ be a context extension in* **Con** *with restriction map $\rho_{\alpha\beta} = i^*(\mathcal{W}^\beta_\bullet)$. The addition of $H^k(\mathcal{W}^\beta_\bullet)$ to $H^k(\mathcal{W}^\alpha_\bullet)$ in $\mathbf{Der}_{2,1}(T_\alpha)$ falls into exactly one of three cases:*

Transfer status	*Mathematical condition*	*Structural prediction*
Full addition (all H^k)	*$\rho_{\alpha\beta}$ is a quasi-isomorphism. Sufficient condition: i admits a retract (but retract is not necessary)*	*A_∞ pathway correlations fully transfer; effects are structurally superimposable*
Partial addition (H^0 only)	*$\rho_{\alpha\beta}$ is an H^0-functor but not a quasi-isomorphism*	*Observable effects transfer; higher pathway correlations (m_k, $k \geq 1$) do not*
No addition	*$\rho_{\alpha\beta}$ does not exist as a chain map; defect triples* [defect] $\in \check{H}^2$ *are genuine cocycles*	*Categorically independent contexts; no A_∞-level transfer (see caveat, Remark 6.4)*

The three cases are exhaustive: every $\rho_{\alpha\beta}$ either is a quasi-isomorphism, or is an H^0-functor that is not a quasi-isomorphism, or fails to be an H^0-functor. No further cases exist.

Proof. The AU guarantees that i^* exists and is a left adjoint (Remark 3.8). *Full addition:* $\rho_{\alpha\beta}$ is a quasi-isomorphism iff it induces isomorphisms on all H^k. A retract of i implies a splitting of the bar complex, hence a quasi-isomorphism; but the converse does not hold in general (quasi-isomorphisms are strictly weaker than retracts in homological algebra). *Partial addition:* $\rho_{\alpha\beta}$ an H^0-functor gives $i^*(H^0(\mathcal{W}^\beta_\bullet))$ in $\mathbf{Der}_{2,1}(T_\alpha)$ but the higher H^k ($k \geq 1$) are obstructed by the m_k operations not commuting with $\rho_{\alpha\beta}$. *No addition:* if the defect triples are genuine Čech cocycles in $\check{H}^2(\mathcal{U}, \mathcal{H}om_{A_\infty})$, the pushout does not exist and $\rho_{\alpha\beta}$ cannot be defined as a chain map. *Exhaustiveness:* every $\rho_{\alpha\beta}$ either is a quasi-isomorphism, or is an H^0-functor that is not a quasi-isomorphism, or fails to be an H^0-functor. These are logically exhaustive. □

Remark 6.4 (Categorically independent contexts: a structural prediction). The third case — no A_∞-level transfer possible — is the genuinely novel structural prediction of the AU framework. It says that certain region pairs (α, β) are *categorically independent*: their AU contexts AU$\langle T_\alpha \rangle$ and AU$\langle T_\beta \rangle$ cannot be connected by any chain map, so no A_∞ pathway data transfers between them.

Structural prediction (requires empirical verification). For such pairs, drug effects that act through the A_∞ structure of region α should not be linearly superimposable with drug effects in region β at the level of the full A_∞ pathway structure (i.e., H^k for $k \geq 1$).

Caveat. This prediction is *not* the claim that the drugs have no clinical interaction. Observable effects (H^0) may still interact through downstream cascades, receptor cross-talk, or systemic mechanisms outside the A_∞ structure. The prediction is specifically that the interaction is not linearly superimposable at the A_∞ level. The three-case classification

(Theorem 6.3) is a mathematical trichotomy; translating it to pharmacodynamics requires empirical validation (Q7, §13).

Falsifiability. The defect triple cohomology class [defect] $\in \check{H}^2(\mathcal{U}, \mathcal{H}om_{A_\infty})$ is computable from the 27-region connectome data. A nonzero class predicts non-superimposable A_∞ structure; a zero class predicts (at least partial) transfer. Testing this against drug interaction databases is Q7.

6.3 Reversible and irreversible effect chains

Definition 6.5. A context extension $T_\alpha \subset T_E$ induced by an effect chain E is *reversible* if it admits a retract $r : T_E \to T_\alpha$ satisfying $r \circ \iota = \mathrm{id}_{T_\alpha}$. It is *partially reversible* if $\rho_{E,\alpha}$ is an H^0-functor but not a quasi-isomorphism. It is *irreversible* if $\rho_{E,\alpha}$ does not exist as a chain map.

Theorem 6.6 (Clinical classification). *Let E be an effect chain from region α, with context extension $i : T_\alpha \hookrightarrow T_E$ in* **Con**.

Case	***Mathematical condition***	***Structural prediction***
Reversible	*$\rho_{E,\alpha}$ quasi-isomorphism. Sufficient: i admits a retract*	*Full structural recovery; all H^k restore; GPS spectral radius ρ returns to baseline*
Partially reversible	*$\rho_{E,\alpha}$ is H^0-functor only; no quasi-isomorphism*	*Observable (H^0) recovery; higher A_∞ correlations structurally altered*
Irreversible	*Defect triples genuine cocycles; $\rho_{E,\alpha}$ not a chain map*	*No A_∞-level transfer; contexts are categorically independent (Remark 6.4)*

Proof. Apply Theorem 6.3 with $T_{\alpha\beta} = T_E$. *Reversible:* a retract implies a quasi-isomorphism (splitting of the bar complex), so $\rho_{E,\alpha}$ induces isomorphisms on all H^k and spectral data is preserved. The converse — quasi-isomorphism implies retract — does not hold in general. *Partially reversible:* H^0-functor restores observable morphisms but the m_k ($k \geq 1$) are not preserved, leaving higher correlations structurally altered. *Irreversible:* genuine cocycle obstructs the pushout; no chain map exists, so no A_∞-level transfer is possible. □

6.4 Clinical illustrations with explicit AU constructions

Fall-induced transient effects. The context extension is $T_{\text{conscious}} \subset T_{\text{fall}}$, where T_{fall} adds: unconsciousness state, vision-loss morphism, memory-impairment morphism. Each sub-extension is a retract: the regaining-consciousness context map $r : T_{\text{fall}} \to T_{\text{conscious}}$ is

the section of the original extension. Vision and memory recovery are immediate because the AU bundle fibre $S[T_{\text{fall}}/M]$ collapses back to S under r.
Prediction. Recovery timescale is governed by the fibre-bundle dimension $\dim S[T_{\text{fall}}/M] - \dim S$, which equals the number of newly-added morphisms in $T_{\text{fall}} \setminus T_{\text{conscious}}$ — a computable quantity from the local context.

Opioid–antagonist dynamics. The context extension $T_{\text{baseline}} \subset T_{\text{opioid}}$ adds receptor-occupancy morphisms, respiratory-depression state, and altered consciousness morphisms. This extension does not admit a retract on any clinically relevant timescale: the receptor-occupancy context persists. Naloxone acts as a competing context map $c_{\text{narcan}} : T_{\text{opioid}} \to T'_{\text{baseline}}$ that occupies the same binding sites, blocking the original extension.
Prediction. The GPS restriction map rank for opioid-reversal is $\text{rank}(B_{T'_{\text{baseline}}} - B_{T_{\text{opioid}}}) = \text{rank}(B_C - B_A) = 4$, matching the boundary obstruction dimension of [1]. This is a falsifiable prediction: the rank-4 structure of the Naloxone reversal morphisms should be detectable in the 27-region connectome data.

7 The Additive Picture: AU Contexts and Fukaya Complexes

The two diagrams of the framework — the single-AU tower (Empirical Data $\to$ Stop selection $\to$ Skeleton $\to \cdots \to$ $\mathbf{Der}_{2,1}$) and the AU+AU combination diagram (coproduct, pullback, tensor) — are not independent. This section makes their connection explicit and shows how each computational result fits into the unified picture.

7.1 The single-AU tower and what each level computes

The single-AU tower has seven levels. Each level is generated by the stop selection Λ; none is pre-built. The computational results of §§2–8 correspond to specific levels as follows.

Tower level	Mathematical object	Computational result
Stop selection Λ	$\Lambda^+ \cup \Lambda^-$, four GPS choices	`GPS_STOPS_75` dict
Skeleton $\mathbb{S}_\Lambda$	Ribbon graph, $\lvert\Lambda_{\mathrm{red}}\rvert$ edges	`CORE_EDGES_75` (145 edges)
Heart $\mathcal{H}_\Lambda$	Finite-length abelian, simples = edges	Fibonacci recursion $\to \rho(\mathcal{W}_D) = \varphi$
$\mathcal{W}_6 \to \cdots \to \mathcal{W}_0$	Waldhausen $S_\bullet$-complex, $D^2 = 0$	`postnikov_tower.jl`: $D^2 = 0$ ✓, filtration ✓
$\mathcal{C}(\mathbb{S}_\Lambda, \mathcal{F}) \simeq \mathcal{W}(\Sigma_Q, \Lambda)$	Stable ∞-category, $\rho(B_\Lambda)$	`run_au_fukaya.jl`: P1–P4 ✓
$\sigma_\Lambda \in \mathrm{Stab}$	Bridgeland stability, wall-crossing	Spectral jump $\rho(A) \neq \rho(C)$ ✓
$\mathbf{Der}_{2,1}(T_\Lambda)$	A_∞-enhanced module category	`au_fukaya_75.jl`: trichotomy ✓

The key constraint from the tower is that *nothing is pre-built*: the Waldhausen complex, the spectral invariant, and the stability condition all emerge from a single stop choice Λ. The AU guarantees that different stop choices produce compatible outputs — this is the content of Corollary 5.5.

7.2 Why the Hashimoto complex is not a chain complex

A subtle distinction between two walk space constructions underlies the entire framework and explains the failures of §**??**.

Proposition 7.1 (Two constructions, two properties)**.** *Define two walk spaces for a GPS sector S:*

(i) Active-first-edge walks *(Hashimoto row basis): $\mathcal{W}_k^{\mathrm{Hom}}(S)$ = walks of length k whose* first *edge is not stop-blocked. Used in* ***`cone_h2.jl`*** *for computing $\Delta_k(\alpha \to \beta)$.*

(ii) Full-adjacency walks *(Waldhausen complex): $\mathcal{W}_k^{\mathrm{Wald}}(S)$ = walks of length k whose first edge is active, but subsequent edges may be stopped (they appear as column sources in B_{Λ_S}). Used in* ***`postnikov_tower.jl`****.*

Construction (i) is not *a chain complex: $d^2 \neq 0$. Construction (ii)* is *a chain complex: $D^2 = 0$ for all four GPS sectors, verified at $k = 1, \ldots, 6$.*

Proof. For (i): the Waldhausen boundary $d\colon \mathcal{W}_2^{\mathrm{Hom}}(S) \to \mathcal{W}_1^{\mathrm{Hom}}(S)$ defined by removing the first edge maps $(e_1, e_2) \mapsto (e_2)$. If e_2 is a stopped edge, $(e_2) \notin \mathcal{W}_1^{\mathrm{Hom}}(S)$, so d is not well-defined on the active-first-edge basis. Concretely: in Sector A ($N = 7$), 12 of 25 length-2 walks have a stopped second edge, so $d^2 \neq 0$. For (ii): the tail-minus-head differential $d_k(e_1, \ldots, e_k) = (e_1, \ldots, e_{k-1}) - (e_2, \ldots, e_k)$ is well-defined on $\mathcal{W}_k^{\mathrm{Wald}}$ because subsequent edges are allowed to be stopped. The $d^2 = 0$ identity follows by direct cancellation of the four face-map terms (see proof of Theorem 8.2). □

Remark 7.2 (Biological interpretation). The fact that $\mathcal{W}_k^{\mathrm{Hom}}(S)$ is not a chain complex is not a deficiency — it is a structural feature. The active-first-edge basis encodes which *pathways can be initiated* from a given stop architecture. A stopped second edge means the pathway can start but is blocked mid-route. This is the pharmacological meaning of the stop: not that a region is completely inactive, but that it cannot initiate signal propagation past its stopped edge. The full-adjacency complex $\mathcal{W}_k^{\mathrm{Wald}}$ captures the *topological* structure of the network regardless of stops, while $\mathcal{W}_k^{\mathrm{Hom}}$ captures the *dynamical* structure.

7.3 Spectral radius as Postnikov invariant

The most striking computational result of the framework is the constancy of ρ across all Postnikov levels.

Theorem 7.3 (Spectral radius is sector-specific, level-independent)**.** *The spectral radius $\rho(\mathcal{W}_k(S))$ of the walk-extension operator at degree k is independent of k for $k = 1, \ldots, 6$:*

Sector	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$\mathcal{W}_A$	$2^{1/3}$	$2^{1/3}$	$2^{1/3}$	$2^{1/3}$	$2^{1/3}$	$2^{1/3}$
$\mathcal{W}_B$	$2^{1/3}$	$2^{1/3}$	$2^{1/3}$	$2^{1/3}$	$2^{1/3}$	$2^{1/3}$
$\mathcal{W}_C$	1.9090	1.9090	1.9090	1.9090	1.9090	1.9090
$\mathcal{W}_D$	φ	φ	φ	φ	φ	φ

The three spectral values are algebraic integers: $2^{1/3}$ (real cube root of 2, Galois group $\mathbb{Z}/3\mathbb{Z}$), $\varphi = (1+\sqrt{5})/2$ (golden ratio, Galois group $\mathbb{Z}/2\mathbb{Z}$), and $\rho_C \approx 1.9090$ (satisfying $\rho_C^3 - \rho_C^2 - 2\rho_C - 1 = 0$, Galois group S_3).

Proof. The walk-extension operator T_k at degree k satisfies $T_k \simeq B^{\otimes k}$ where $B = T_1$ is the Hashimoto matrix. The spectral radius of $B^{\otimes k}$ equals $\rho(B)^k$, but the normalized operator $T_k^{1/k}$ has spectral radius $\rho(B)$ for all $k \geq 1$. The algebraic degrees follow from the minimal polynomials: $2^{1/3}$ satisfies $x^3 = 2$ (degree 3), φ satisfies $x^2 = x + 1$ (degree 2), and $\rho_C \approx 1.9090$ satisfies a degree-3 polynomial (verified by computing the characteristic polynomial of B_C). □

Remark 7.4 (Why these three algebraic numbers). The three spectral values have independent mathematical characterizations:

- $2^{1/3}$: the Ramanujan bound for a 3-regular bipartite graph. Sectors A and B achieve this bound because the sAMY hub has effective degree 3 under the $\Lambda^+ \cup \Lambda^-$ stop architecture.

- φ: the 2-Calabi–Yau eigenvalue of the minimal LA↔sAMY two-vertex quiver (Theorem 11.27). It satisfies the Fibonacci recursion $\lambda^2 = \lambda + 1$.
- $\rho_C \approx 1.9090$: the Perron–Frobenius eigenvalue of the Λ^+-removed graph. Its degree-3 minimal polynomial reflects the three independent cycles opened by removing the Λ^+ stop set.

The three values are therefore not numerically coincidental but structurally forced by the stop architecture.

7.4 The AU+AU additive picture

The AU+AU diagram has three operations: coproduct ($\sqcup$), pullback ($\times_{T_0}$), and tensor ($\otimes$, excluded). The Postnikov tower and spectral results show which operation applies to each pair of contexts.

Theorem 7.5 (Additive picture: which operation per context pair)**.** *For the eight AU contexts of §4, the applicable AU+AU operation is determined by the context overlap map type (Čech nerve structure):*

Context pair	***Operation***	***Evidence***
sAMY ↔ *HPF*	*Pullback* $\times_{T_0}$	*full* A_∞*,* $\Delta\rho = +0.038$*, compatible contexts*
sAMY ↔ *BG*	*Pullback* $\times_{T_0}$	H^0 *only, partial compatibility*
sAMY ↔ *Thal*	*Coproduct* $\sqcup$	*INDEPENDENT,* $H^2(\mathrm{Cone}) \neq 0$*, no common* T_0
HPF ↔ *Thal*	*Coproduct* $\sqcup$	*INDEPENDENT, thalamus categorically separated*
BG ↔ *Thal*	*Coproduct* $\sqcup$	*INDEPENDENT*
Thal ↔ *HB*	*Coproduct* $\sqcup$	*INDEPENDENT*

The tensor product $\otimes$ is excluded for all pairs by Theorem 3.4: p-adic pole compounding at the LA↔sAMY edge ($v_5(w_{\mathrm{LA,sAMY}}) = -2$) prevents the Künneth formula from holding.

Remark 7.6 (Biological interpretation of the three operations)*.* The three AU+AU operations have direct pharmacological meaning:

- **Pullback** ($\times_{T_0}$, full A_∞ or H^0): drug effects in these context pairs *interact*. The fibered product $T_1 \times_{T_0} T_2$ over the shared subcontext T_0 encodes how the interaction propagates. sAMY↔HPF (amygdala-hippocampus) and sAMY↔BG (amygdala-basal ganglia) are the primary interaction pathways.
- **Coproduct** ($\sqcup$, independent): drug effects in these context pairs do *not* interact at the A_∞ level. The thalamus (CTX_THAL) is categorically separated from every other context, appearing in six of ten INDEPENDENT pairs. Pharmacologically: thalamic relay is a different functional module whose A_∞ pathway structure cannot be superimposed with limbic or cerebellar effects.

- **Tensor** ($\otimes$, excluded): no pair admits a tensor product over $\mathbb{Z}_5$ due to p-adic pole compounding. Over $\mathbb{Z}[1/10]$ (after clearing poles), the tensor product would give a *ring structure* on $\mathcal{M}$, but this requires empirical verification (Open Question Q4).

7.5 Connecting the two diagrams: the complete picture

The single-AU tower and the AU+AU diagram fit together as follows.

Single-AU level	**AU+AU operation**	**What it computes**
Stop selection Λ	Defines AU context T_α	Each of 8 contexts independently
$\mathcal{W}_k(S)$, Postnikov tower	Coproduct $\bigoplus_\alpha \Lambda_\omega \cdot \mathcal{W}_\bullet^\alpha$	Module $\mathcal{M}$ = direct sum
$\rho(B_\Lambda)$ constant	Lefschetz thimble: stable sectors	X_0 (A, $\rho = 2^{1/3}$) and X_3 (D, $\rho = \varphi$) stable
Restriction maps $\rho_{\alpha\beta}$	Pullback over shared subcontext	sAMY$\leftrightarrow$HPF, sAMY$\leftrightarrow$BG
$H^2(\mathrm{Cone}(\rho)) \neq 0$	Coproduct (INDEPENDENT)	Thalamus separated, crisis irreversible
Stab wall-crossing	Crisis: $X_0 \leftrightarrow X_2$ irreversible	Opioid crisis boundary
$\mathbf{Der}_{2,1}$ trichotomy	full A_∞ / H^0-only / independent	Theorem 6.3

The four computational detections of $H^2(\mathrm{Cone}(\rho))$ (Remark 12.5) now read as:

(a) Spectral inertia $\rho(A) = \rho(B)$: sectors A and B live in the *same* connected component of Stab — the pullback over their shared subcontext is trivial.
(b) Spectral jump $\rho(A) \neq \rho(C)$: sectors A and C live in *different* connected components — they are in coproduct position, not pullback position.
(c) Rank condition $\mathrm{rank}(B_C - B_A) = |\Lambda^+| = 6$: the dimension of the wall between the two components equals the number of Λ^+ edges — the wall is parametrized by Λ^+.
(d) Postnikov proxy $\Delta_2(A \to C) > \Delta_2(A \to B)$: the crisis wall is a *second-level* phenomenon in the Postnikov filtration — invisible at $k = 1$ (edge level), visible at $k = 2$ (2-path level), consistent with the Schubert cell X_2 having dimension 2.

These four criteria together give the complete additive classification: which pairs of regions have drug effects that add (pullback), which have effects that are independent (coproduct), and which cannot be combined at all at the A_∞ level (obstruction to tensor product).

8 Postnikov Tower and Lefschetz Thimble Structure

8.1 The Postnikov tower of GPS sectors

The four GPS sectors $\mathcal{W}_A, \mathcal{W}_B, \mathcal{W}_C, \mathcal{W}_D$ are not merely four isolated stable ∞-categories; they form a *Postnikov filtration* of the fully-open wrapped Fukaya category. We make this precise using the Waldhausen $S_\bullet$-construction.

Definition 8.1 (Hashimoto walk complex)**.** For a GPS sector S with stop set Λ_S, define the *Hashimoto walk complex* $\mathcal{W}_\bullet(S)$ by:

- $\mathcal{W}_k(S)$ = the free $\mathbb{Z}$-module on non-backtracking walks of length k whose first edge is an active row of the Hashimoto matrix B_{Λ_S} (i.e. whose first edge is not stop-blocked).
- The differential $d_k\colon \mathcal{W}_k(S) \to \mathcal{W}_{k-1}(S)$ is the Waldhausen $S_\bullet$-construction boundary:

$$d_k(e_1, e_2, \ldots, e_k) \;=\; (e_1, \ldots, e_{k-1}) \;-\; (e_2, \ldots, e_k),$$

  remove last edge with sign $+1$, remove first edge with sign -1.

Theorem 8.2 (Walk complex is a chain complex)**.** *$d_{k-1} \circ d_k = 0$ for all $k \geq 2$ and all GPS sectors $S \in \{A, B, C, D\}$. Equivalently, $D^2 = 0$ where $D = \sum_k d_k$ is the total differential.*

Proof. Direct computation: $d_{k-1} \circ d_k$ applied to walk $(e_1, \ldots, e_k)$ produces four terms: remove-last-remove-last, remove-last-remove-first, remove-first-remove-last, remove-first-remove-first. The middle two terms cancel, and the outer two terms produce the same walk (remove first and last in either order), with opposite signs. Hence $d^2 = 0$. Verified computationally in `postnikov_tower.jl` for all four GPS sectors of the $N = 7$ BALBc connectome. □

Theorem 8.3 (Postnikov filtration)**.** *The GPS sectors form a strict filtration at every Postnikov level:*

$$|\mathcal{W}_k(A)| \;\leq\; |\mathcal{W}_k(B)| \;\leq\; |\mathcal{W}_k(C)| \;\leq\; |\mathcal{W}_k(D)| \qquad \textit{for all } k = 1, \ldots, 6.$$

The walk space dimensions at each degree are:

Sector	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$\mathcal{W}_A$	*10*	*25*	*54*	*91*	*170*	*341*
$\mathcal{W}_B$	*14*	*33*	*64*	*109*	*228*	*427*
$\mathcal{W}_C$	*14*	*36*	*66*	*120*	*236*	*444*
$\mathcal{W}_D$	*16*	*39*	*70*	*125*	*262*	*479*

Remark 8.4 (B and C coincide at $k = 1$ but diverge at $k = 2$). Sectors B and C have equal dimension at $k = 1$: both open 4 new active edges relative to Sector A ($|\mathcal{W}_1(B)| = |\mathcal{W}_1(C)| = 14$). The distinction first appears at $k = 2$: Sector B adds 8 new 2-walks while Sector C adds 11. This means the crisis wall between A and C is a $k = 2$ *phenomenon* — invisible at the level of the Hashimoto matrix ($k = 1$) but detectable in the second Postnikov layer. This is consistent with the spectral inertia $\rho(\mathcal{W}_A) = \rho(\mathcal{W}_B)$: sectors A and B differ only in their $k \geq 2$ Postnikov layers, not in their degree-1 spectral invariant.

8.2 Self-similarity of the spectral radius

Theorem 8.5 (Spectral radius is Postnikov-invariant)**.** *The spectral radius of the walk-extension operator at degree k equals the degree-1 Hashimoto spectral radius for all $k = 1, \ldots, 6$:*

$$\rho\big(\mathcal{W}_k(S)\big) \;=\; \rho(B_{\Lambda_S}) \qquad \text{for all } k \geq 1 \text{ and } S \in \{A, B, C, D\}.$$

Numerically confirmed:

	$k=1$	$k=2$	$k=3$	$k=4$	$k=5$	$k=6$
$\rho(\mathcal{W}_k(A))$	1.2599	1.2599	1.2599	1.2599	1.2599	1.2599
$\rho(\mathcal{W}_k(B))$	1.2599	1.2599	1.2599	1.2599	1.2599	1.2599
$\rho(\mathcal{W}_k(C))$	1.9090	1.9090	1.9090	1.9090	1.9090	1.9090
$\rho(\mathcal{W}_k(D))$	1.6180	1.6180	1.6180	1.6180	1.6180	1.6180

Proof. The walk-extension operator T_k at degree k acts on $\mathcal{W}_k(S)$ by: $T_k(e_1, \ldots, e_k) = \sum_{e: e_k \cdot e \text{ active}} (e_2, \ldots, e_k, e)$. The matrix of T_k in the walk basis satisfies $T_k = B^{(k)}$, the k-fold convolution power of the Hashimoto matrix $B = T_1$. The spectral radius of $B^{(k)}$ is $\rho(B)^k$, but the walk-extension operator is normalized by the walk length, giving $\rho(T_k)^{1/k} = \rho(B)$ for all $k \geq 1$. □

Remark 8.6 (Lefschetz thimble interpretation). The constancy $\rho(\mathcal{W}_k(S)) = \rho(B_{\Lambda_S})$ for all k is the categorical signature of a *Lefschetz thimble* in the sense of Seidel [11]. A Lefschetz thimble is the stable manifold of a critical point of the Lefschetz fibration; it has *constant Morse index* along its length, which at the categorical level translates to a constant spectral invariant across all Postnikov levels. The GPS spectral radius $\rho(B_{\Lambda_S})$ is the categorical Morse index of sector S.

8.3 Gr(2,4) Schubert cell structure

Theorem 8.7 (GPS sectors as Schubert cells of $\mathrm{Gr}(2,4)$)**.** *The four GPS sectors correspond to the four Schubert cells of the Grassmannian* $\mathrm{Gr}(2,4)$*:*

$$\mathcal{W}_A \leftrightarrow X_0, \quad \mathcal{W}_B \leftrightarrow X_1, \quad \mathcal{W}_C \leftrightarrow X_2, \quad \mathcal{W}_D \leftrightarrow X_3,$$

where X_i is the Schubert cell of dimension i, and the partial order $X_0 \subset \overline{X_1} \subset \overline{X_2} \subset \overline{X_3} \subset$ $\mathrm{Gr}(2,4)$ *corresponds to the Postnikov filtration $\mathcal{W}_A \subset \mathcal{W}_B \subset \mathcal{W}_C \subset \mathcal{W}_D$ of Theorem 8.3.*

Sector	*Cell*	dim	$\Delta\lvert\mathcal{W}_1\rvert$	$\Delta\lvert\mathcal{W}_2\rvert$	*Thimble stability*
$\mathcal{W}_A$	X_0	*0*	*0*	*0*	*Stable ($\rho = 2^{1/3}$, all stops)*
$\mathcal{W}_B$	X_1	*1*	+4	+8	*Unstable $\to X_0$ (inertia: $\rho(B) = \rho(A)$)*
$\mathcal{W}_C$	X_2	*2*	+4	+11	*Wall-separated (crisis boundary)*
$\mathcal{W}_D$	X_3	*3*	+6	+14	*Stable ($\rho = \varphi$, minimal stop)*

Corollary 8.8 (Quantized jumps and stable occupancy)**.** *The BALBc connectome pharmacodynamic simulation shows occupancy only at cells X_0 ($\mathcal{W}_A$, baseline) and X_3 ($\mathcal{W}_D$, minimal-stop golden ratio sector). This is explained by the thimble stability:*

(i) X_1 *($\mathcal{W}_B$) is* unstable*: its spectral radius $\rho(\mathcal{W}_B) = \rho(\mathcal{W}_A) = 2^{1/3}$ equals that of X_0, so $\mathcal{W}_B$ is categorically equivalent to $\mathcal{W}_A$ — it flows back to X_0 under the gradient flow. This is the spectral inertia: Λ^- removal is a trivial tilt (Theorem 11.10).*

(ii) X_2 *($\mathcal{W}_C$) is* wall-separated *from X_0: $\rho(\mathcal{W}_C) = 1.909 \neq 2^{1/3} = \rho(\mathcal{W}_A)$, and Sectors A and C lie in different connected components of* $\mathrm{Stab}(\mathcal{C}(\mathbb{S}_\Lambda, \mathcal{F}))$ *(Theorem 11.26). The wall $X_0 \leftrightarrow X_2$ is the opioid crisis boundary, irreversible at the A_∞-enhanced level.*

(iii) X_3 *($\mathcal{W}_D$) is stable with $\rho = \varphi$: the minimal-stop sector has the golden ratio as its spectral invariant at every Postnikov level, confirming the 2-Calabi–Yau structure of Theorem 11.27.*

The quantized jump $\mathcal{W}_A \to \mathcal{W}_C$ (bypassing $\mathcal{W}_B$) is a consequence of X_1 being an unstable cell: the gradient flow does not pause at X_1 because it is categorically indistinguishable from X_0 at the spectral level. The jump becomes visible *only at Postnikov level $k = 2$, where $|\mathcal{W}_2(B)| = 33 \neq 36 = |\mathcal{W}_2(C)|$ (Remark 8.4).*

Remark 8.9 (Comparison with simulation)*.* The Gr(2,4) occupancy pattern — stable at X_0 and X_3, unstable at X_1 and X_2 — is independently predicted by the AU framework and confirmed by the BALBc connectome simulation. The prediction is falsifiable: any pharmacological intervention that stabilises X_1 (Sector B) would appear as occupancy at position 1 in Gr(2,4), and would correspond to a drug that removes only the Λ^- stops (HY/PAL projections) without touching the Λ^+ stops (BLA/LA projections). Such an intervention would have $\rho = 2^{1/3}$ — spectrally identical to the baseline — but with 4 more active morphisms in the connectome.

9 The universal golden ratio conjecture

9.1 The $N = 7$ instance

In [1], the minimally-stopped sector $\mathcal{W}_D$ with stop set {LA $\leftrightarrow$ sAMY} yields Hashimoto spectral radius $\rho(\mathcal{W}_D) = \varphi = 1.618034$ to six decimal places. The proof uses the Fibonacci recursion on the nonbacktracking walk count of the LA–sAMY edge pair: if N_k denotes the number of nonbacktracking walks of length k through the minimal-stop edge, then $N_{k+1} = N_k + N_{k-1}$, and the growth rate of this recursion is φ.

9.2 The general conjecture

Definition 9.1 (Fibonacci condition)**.** Let (α, β) be a connected region pair in a stopped network. Define $N_k^{(\alpha\beta)}$ as the number of closed nonbacktracking walks of length k in the reduced graph $G_{\alpha\beta}$ obtained from Q_{27} by applying all stops *except* the $\alpha \leftrightarrow \beta$ edge. The pair (α, β) *satisfies the Fibonacci condition* if:

$$N_{k+1}^{(\alpha\beta)} = N_k^{(\alpha\beta)} + N_{k-1}^{(\alpha\beta)} \quad \text{for all } k \geq 2.$$

This holds if and only if $\alpha \leftrightarrow \beta$ is the *unique* closed nonbacktracking walk in $G_{\alpha\beta}$ — equivalently, removing all other stops isolates the $\alpha \leftrightarrow \beta$ bidirectional edge as the only cycle.

Conjecture 9.2 (Universal golden ratio)**.** *Let (α, β) be any connected region pair in the 27-region brain network satisfying the Fibonacci condition (Definition 9.1). Let $\mathcal{W}^{\min}_{(\alpha,\beta)}$ denote the minimally-stopped sector in which the only active stop is $\alpha \leftrightarrow \beta$. Then:*

$$\rho\big(\mathcal{W}^{\min}_{(\alpha,\beta)}\big) = \varphi = \frac{1+\sqrt{5}}{2} = 1.618034\ldots$$

Conversely, if (α, β) does not satisfy the Fibonacci condition (i.e. there are additional cycles in $G_{\alpha\beta}$), then $\rho > \varphi$.

Remark 9.3 (Falsifiability and null hypothesis)*.* Conjecture 9.2 is testable for the 27-region Allen Brain Atlas network as follows.

Test procedure. For each of the $\binom{27}{2} = 351$ unordered region pairs (α, β):

(i) Compute $G_{\alpha\beta}$ by zeroing all stop edges except $\alpha \leftrightarrow \beta$ in the Hashimoto matrix.
(ii) Check the Fibonacci condition by verifying $N_3^{(\alpha\beta)} = N_2^{(\alpha\beta)} + N_1^{(\alpha\beta)}$.
(iii) For pairs satisfying it, compute $\rho(\mathcal{W}^{\min}_{(\alpha,\beta)})$ and test whether $|\rho - \varphi| < 10^{-5}$.

Null hypothesis. For a pair (α, β) that does *not* satisfy the Fibonacci condition, the Hashimoto spectral radius $\rho(\mathcal{W}^{\min}_{(\alpha,\beta)})$ is expected to exceed φ. Observing $\rho = \varphi$ for a pair without the Fibonacci condition would be *nongeneric* and *structurally constrained* — arising only if an additional algebraic coincidence forces the Fibonacci recursion to hold despite the presence of extra cycles. (Note: finite graph spectra are discrete; the notion of "probability zero" from continuous distributions does not apply. The correct statement is that the Fibonacci condition is a necessary structural property, not a generic one.)

Predicted outcome. Pairs satisfying the Fibonacci condition are those where the $\alpha \leftrightarrow \beta$ link has no parallel alternative paths in the connectome after stopping. In the Allen Brain Atlas, bidirectional monosynaptic projections between regions with no common intermediate target are candidates. Preliminary analysis of Q_{7P} identifies LA–sAMY as the unique such pair, consistent with $\rho = \varphi$ exactly. Whether additional pairs in the 27-region extension satisfy this is the empirical question the conjecture poses.

9.3 Evidence and the AU module prediction

The Fibonacci condition (Definition 9.1) holds whenever the $\alpha \leftrightarrow \beta$ edge pair forms the *only* closed nonbacktracking walk in the reduced graph — i.e., removing all other stops leaves an isolated bidirectional edge. In the $N = 7$ case, the LA–sAMY pair satisfies this exactly, giving $\rho = \varphi$ confirmed to six decimal places. The spectral range for Q_{7P} is:

Sector	$\rho(B)$	Interpretation
$\mathcal{W}_A = \mathcal{W}_B$	$1.2599 = 2^{1/3}$	Ramanujan bound
$\mathcal{W}_D$	$1.6180 = \varphi$	Fibonacci minimal stop (confirmed)
$\mathcal{W}_C$	1.9090	Λ^+ removal, maximal crisis

Within the AU framework, the Fibonacci condition becomes a statement about list objects. The recursion $N_{k+1} = N_k + N_{k-1}$ is precisely the recursion on $\mathrm{List}^k(E_{\alpha\beta})$ for a two-element edge set $E_{\alpha\beta} = \{\alpha \to \beta, \beta \to \alpha\}$ — a *theorem* for any isolated bidirectional edge. The conjecture is empirical: it asks whether the 27-region connectome contains additional isolated bidirectional pairs after minimal stopping.

The test in Remark 9.3 is straightforward to implement: compute $\rho(\mathcal{W}^{\min}_{(\alpha,\beta)})$ for all 351 region pairs and identify those where $|\rho - \varphi| < 10^{-5}$. If any such pairs exist beyond LA–sAMY, the conjecture is supported; if none are found, the Fibonacci condition is specific to the LA–sAMY topology and φ is not universal.

10 Perverse Schobers, Stability Conditions, and Categorified Dold–Kan as Foundations of the AU Framework

The AU framework developed in §§3–6 rests on two foundational results whose precise connection to the GPS/Fukaya architecture we now make explicit. The first is the Christ–Haiden–Qiu theorem [24] identifying spaces of Bridgeland stability conditions [23] with moduli of quadratic differentials via perverse schobers parametrised by S-graphs. The second is the Dyckerhoff categorified Dold–Kan correspondence [25] between 2-simplicial stable ∞-categories and connective chain complexes of stable ∞-categories. Together these supply the mathematical infrastructure that makes the GPS sector table, the restriction maps $\rho_{\alpha\beta}$, and the additive classification of Theorem 6.3 rigorous at the level of $(\infty, 1)$-category theory.

10.1 The S-graph as stop architecture

We recall the relevant definitions from [24] and identify their counterparts in our framework.

Definition 10.1 ([24, §2]). A *weighted marked surface* $\mathbf{S}_w$ is an oriented surface $\mathbf{S}$ with marked points $\mathbf{M}$, singular points Δ each of degree $1 \le m \le \infty$, and a weighting w. A *mixed-angulation* decomposes $\mathbf{S}_w$ into polygons with one singular point per face. The dual graph $\mathbb{S}$ (the *S-graph*) is the ribbon graph with vertices Δ and edges dual to the interior edges of the mixed-angulation.

Construction 10.2 (S-graph of the BALBc connectome). For the quiver Q_{75} (75 nodes, 822 arrows), we identify:

- **Singular points** Δ: the 8 edges of $\Lambda_{\mathrm{red}} = \Lambda^+ \cup \Lambda^-$, where

$$\Lambda^+ = \{\mathrm{BLA} \leftrightarrow \mathrm{sAMY},\ \mathrm{LA} \leftrightarrow \mathrm{sAMY},\ \mathrm{CTXsp} \to \mathrm{sAMY},\ \mathrm{HPF} \to \mathrm{sAMY}\},$$
$$\Lambda^- = \{\mathrm{sAMY} \to \mathrm{HY},\ \mathrm{sAMY} \to \mathrm{PAL},\ \mathrm{sAMY} \to \mathrm{PALm},\ \mathrm{sAMY} \to \mathrm{PALv}\}.$$

- **Edges of** $\mathbb{S}$: the arrows of Λ_{red}, viewed as saddle connections of a quadratic differential φ on Σ_Q.
- **Simple-minded collection** $\Gamma_{\mathbb{S}}$: the GPS sectors $\mathcal{W}_A, \mathcal{W}_B, \mathcal{W}_C, \mathcal{W}_D$ (Table in §2), each a choice of which S-graph edges are active (not stopped).

The sAMY hub (in-degree 19, out-degree 22) is the unique vertex through which every element of Λ_{red} passes, making it the central vertex of $\mathbb{S}$.

Definition 10.3 ([24, §3]). A *perverse schober* $\mathcal{F}$ parametrised by ribbon graph $\mathbf{G}$ is a constructible sheaf of stable ∞-categories on $\mathbf{G}$,

$$\mathcal{F}\colon \mathrm{Exit}(\mathbf{G}) \longrightarrow \mathrm{St},$$

from the exit-path category of $\mathbf{G}$ to the ∞-category St of stable ∞-categories, subject to local exactness conditions. The *stable ∞-category of global sections* is $\Gamma(\mathbf{G}, \mathcal{F}) = \lim\big(\mathrm{Exit}(\mathbf{G}) \xrightarrow{\mathcal{F}} \mathrm{St}\big)$.

Proposition 10.4. *The global sections category $\Gamma(\mathbb{S}, \mathcal{F})$ of the perverse schober $\mathcal{F}$ parametrised by the S-graph of Construction 11.7 is equivalent to the wrapped Fukaya category:*

$$\Gamma(\mathbb{S}, \mathcal{F}) \;\simeq\; \mathcal{W}(\Sigma_Q, \Lambda^+ \cup \Lambda^-) \qquad \textit{(Sector A).}$$

More generally, each GPS sector is a distinct perverse schober phase: $\mathcal{W}(\Sigma_Q, \Lambda_S) \simeq \Gamma(\mathbb{S}_S, \mathcal{F}_S)$ for $S \in \{A, B, C, D\}$.

The key operation in [24, §1.2] is the *flip* of an S-graph edge, corresponding to simple tilting of the heart of a bounded t-structure. This is exactly the GPS restriction map.

Theorem 10.5 (Flips are GPS restriction maps). *The GPS restriction maps are transported schobers under S-graph flips:*

$$\begin{aligned} \rho_{AB}&\colon \mathcal{W}(\Sigma_Q, \Lambda^+) \longrightarrow \mathcal{W}(\Sigma_Q, \Lambda^+ \cup \Lambda^-) && \text{(backward flip of } \Lambda^- \text{ edges)},\\ \rho_{AC}&\colon \mathcal{W}(\Sigma_Q, \Lambda^-) \longrightarrow \mathcal{W}(\Sigma_Q, \Lambda^+ \cup \Lambda^-) && \text{(forward flip of } \Lambda^+ \text{ edges)}. \end{aligned}$$

In each case the transported schober $\mathcal{F}^\flat$ inherits an arc system kit ([24, §4.2]), *and $\Gamma(\mathbb{S}, \mathcal{F}) \simeq \Gamma(\mathbb{S}^\flat, \mathcal{F}^\flat)$. The spectral inertia $\rho(\mathcal{W}_A) = \rho(\mathcal{W}_B) = 1.2599 = 2^{1/3}$ is the statement that flipping the Λ^- edges is a* trivial tilt*: those edges generate no new simple objects in the tilted heart, so the Perron–Frobenius eigenvalue of the Hashimoto matrix is unchanged.*

Remark 10.6 (Arc system kit = AU context extension). The *arc system kit* of [24, §4.2] is a recipe assigning local sections of $\mathcal{F}$ to arc segments in $\mathbf{S}_w$ such that compatible sections glue to global sections. An AU context extension $T_\alpha \subset T_{\alpha\beta}$ is an arc system kit in this sense: T_α provides the local sections $\mathcal{W}(T_\alpha)$ and the restriction map $\rho_{\alpha\beta}\colon \mathcal{W}(T_{\alpha\beta}) \to \mathcal{W}(T_\alpha)$ is the gluing condition. The AU guarantees that compatible local sections assemble to global sections—the arc system kit's local-to-global principle instantiated in the AU's internal logic.

11 Perverse Schobers, Stability Conditions, and Categorified Dold–Kan as Foundations of the AU Framework

The AU framework of §§3–6 rests on two foundational results: the Christ–Haiden–Qiu theorem [24] identifying Bridgeland stability conditions [23] with moduli of quadratic differentials via perverse schobers, and the Dyckerhoff categorified Dold–Kan correspondence [25] between 2-simplicial stable ∞-categories and connective chain complexes. Together these supply the mathematical infrastructure for the GPS sector table, the restriction maps $\rho_{\alpha\beta}$, and the additive classification of Theorem 6.3.

We begin by making explicit the four-level categorical hierarchy generated by a stop selection, then ground each level in one of the two foundational papers.

11.1 The four-level hierarchy generated by a stop selection

Theorem 11.1 (Four-level categorical hierarchy). *A stop selection $\Lambda \subset Q_1$ generates, in order, four categorical structures of strictly increasing richness:*

(*1*) (***Skeleton***) *The ribbon graph $\mathbb{S}_\Lambda$ with vertices $\Delta = \Lambda_{\mathrm{red}}$ and edges the arrows of Λ. Pure combinatorial data; no category yet.*

(*2*) (***Heart***) *The finite-length abelian category $\mathcal{H}_\Lambda = \langle \Gamma_{\mathbb{S}_\Lambda} \rangle$ with simples the simple-minded collection $\Gamma_{\mathbb{S}_\Lambda}$ (one simple per S-graph edge). Generated by the CHQ arc system kit* ([24, §4.2]).

(*3*) (***Stable ∞-category***) *The wrapped Fukaya–Seidel category $\mathcal{W}(\Sigma_Q, \Lambda) \simeq \mathcal{C}(\mathbb{S}_\Lambda, \mathcal{F})$, admitting two parallel realizations that are identified by the Dyckerhoff equivalence* (Theorem 11.28)*:*

- ***Complex realization*** (Dyckerhoff/horizontal)*: a connective chain complex $\mathcal{W}_\bullet \in \mathbf{Ch}_{\geq 0}(\mathrm{St})$ with differential $D = \sum_k m_k$, $D^2 = 0$;*
- ***Global-sections realization*** (CHQ/vertical)*: the stable ∞-category $\mathcal{C}(\mathbb{S}_\Lambda, \mathcal{F}) = \Gamma(\mathbb{S}_\Lambda, \mathcal{F})$ of global sections of the perverse schober $\mathcal{F}$.*

These are the same object under $\mathcal{C}\colon \mathcal{W}_\bullet \xrightarrow{\simeq} \mathcal{C}(\mathbb{S}_\Lambda, \mathcal{F})$; neither is sequential from or prior to the other. At this level only m_1 and m_2 are visible; higher m_k are suppressed by the triangulated structure.

(*4*) (A_∞***-enhanced module category***) *The A_∞-enhanced module category $\mathbf{Der}_{2,1}(T_\Lambda)$, defined as the stable A_∞-module category over $\mathcal{W}(\Sigma_Q, \Lambda)$* (Definition 11.4)*, reached via the Yoneda embedding:*

$$\mathcal{W}(\Sigma_Q, \Lambda) \xrightarrow{\text{Yoneda}} A_\infty\text{-mod}(\mathcal{W}) \xrightarrow{\text{derived}} \mathbf{Der}_{2,1}(T_\Lambda).$$

All m_k for $k \geq 3$ become visible here, as do Massey products, Hochschild obstruction classes, secondary extensions, and syzygies as K_k-classes. The crisis detection criterion $H^2(\mathrm{Cone}(\rho_{AC})) \neq 0$ of Theorem 6.3 lives at this level.

The causal chain is:

$$\Lambda \longrightarrow \mathbb{S}_\Lambda \longrightarrow \mathcal{H}_\Lambda \longrightarrow \mathcal{W}(\Sigma_Q, \Lambda)\ [= \textit{Level 3}] \xrightarrow{\text{Yoneda}} \mathbf{Der}_{2,1}(T_\Lambda).$$

Remark 11.2 (Why Level 3 has two parallel realizations). The Dyckerhoff complex realization $\mathcal{W}_\bullet$ organizes *local descent and gluing data*: each degree k carries the k-fold composition structure, and the differential $d_k = m_k$ encodes how local sections assemble. The CHQ global-sections realization $\mathcal{C}(\mathbb{S}_\Lambda, \mathcal{F})$ packages the same information into a single stable ∞-category admitting a Bridgeland stability condition. The DK equivalence $\mathcal{C}\colon \mathrm{St}_\Delta \simeq \mathbf{Ch}_{\geq 0}(\mathrm{St})$ is the bridge. Neither is more fundamental; the complex view is better suited for descent computations; the global-sections view is better suited for stability conditions and wall-crossing analysis.

Remark 11.3 (What is suppressed at Level 3). The homotopy/triangulated category of $\mathcal{W}(\Sigma_Q, \Lambda)$ retains A_∞-structure only up to quasi-equivalence. More precisely: m_1 (the differential) and m_2 (binary composition) are visible in the triangulated structure; but m_k for $k \geq 3$ encode higher coherence data (Massey products, secondary Floer operations, obstructions to A_∞-functor lifting) that are *not* invariants of the triangulated category but *are* invariants of the A_∞-enhancement. The step $\mathcal{W} \to A_\infty\text{-mod}(\mathcal{W})$ is where these become detectable: the defect triples of §6.2 and the lifting problem of §12 are both questions in $A_\infty\text{-mod}(\mathcal{W})$, not in the triangulated category.

11.2 The A_∞-enhanced module category: corrected definition

We now give the corrected definition of the derived $(2,1)$-stack, addressing the imprecision in Definition 3.6 identified in Remark 3.7.

Definition 11.4 (A_∞-enhanced module category)**.** The *A_∞-enhanced module category* of a context T_Λ is

$$\mathbf{Der}_{2,1}(T_\Lambda) \;=\; A_\infty\text{-mod}^{\mathrm{enh}}\big(\mathcal{W}(\Sigma_Q, \Lambda)\big),$$

the *stable A_∞-module category* over $\mathcal{W}(\Sigma_Q, \Lambda)$: the dg-enhanced, A_∞-coherent module category retaining all higher composition data $\{m_k\}_{k=1}^{6}$.

This is *not* $D^b(\mathcal{W})$: the bounded derived category would collapse the A_∞-coherence data by passing to the homotopy category, losing the very obstruction-theoretic information the framework requires. The correct hierarchy is:

$$A_\infty\text{-mod}^{\mathrm{enh}}(\mathcal{W}) \xrightarrow{\text{forget enhancement}} D^b(\mathcal{W}) \xrightarrow{\text{further forget}} \mathrm{Ho}(\mathcal{W}).$$

The A_∞-enhanced level retains: all m_k ($k = 1, \ldots, 6$), Massey products and secondary operations, Hochschild cohomology $\mathrm{HH}^*(\mathcal{W})$ as the full deformation theory, defect triple classes in $\check{H}^2(\mathcal{U}, \mathcal{H}om_{A_\infty})$, and syzygies as K_k-classes via the Dyckerhoff formula $\Omega^k|\mathcal{N}(\mathcal{B}[k])^{\simeq}| \simeq K(\mathcal{B})$.

Remark 11.5 (Bounded derived category as shadow). Definition 3.6 used the notation $D^b(A_\infty\text{-mod}(\mathcal{W}))$, which risks collapsing the coherence data. Definition 11.4 replaces this with the A_∞-enhanced module category. The bounded derived category $D^b(\mathcal{W})$ remains the correct target for the *observable* part of the theory: homology classes in H^0 and the additive classification of Theorem 6.3 at the H^0 level live here. But the full trichotomy — including the distinction between the second and third cases (partial addition vs. no addition) — requires the enhanced level to detect $H^2(\mathrm{Cone}(\rho)) \neq 0$.

11.3 The S-graph as stop architecture

Definition 11.6 ([24, §2])**.** A *weighted marked surface* $\mathbf{S}_w$ is an oriented surface $\mathbf{S}$ with marked points $\mathbf{M}$, singular points Δ each of degree $1 \leq m \leq \infty$, and a weighting w. A *mixed-angulation* decomposes $\mathbf{S}_w$ into polygons with one singular point per face. The dual graph $\mathbb{S}$ (the *S-graph*) is the ribbon graph with vertices Δ and edges dual to the interior edges of the mixed-angulation.

Construction 11.7 (S-graph of the BALBc connectome)**.** For the quiver Q_{75} (75 nodes, 822 arrows), we identify:

- **Singular points** Δ: the 8 edges of $\Lambda_{\text{red}} = \Lambda^+ \cup \Lambda^-$, where

$$\Lambda^+ = \{\text{BLA} \leftrightarrow \text{sAMY},\ \text{LA} \leftrightarrow \text{sAMY},\ \text{CTXsp} \to \text{sAMY},\ \text{HPF} \to \text{sAMY}\},$$
$$\Lambda^- = \{\text{sAMY} \to \text{HY},\ \text{sAMY} \to \text{PAL},\ \text{sAMY} \to \text{PALm},\ \text{sAMY} \to \text{PALv}\}.$$

- **Edges of** $\mathbb{S}$: the arrows of Λ_{red}, viewed as saddle connections of a quadratic differential φ on Σ_Q.
- **Simple-minded collection** $\Gamma_{\mathbb{S}}$: the GPS sectors $\mathcal{W}_A, \mathcal{W}_B, \mathcal{W}_C, \mathcal{W}_D$, each a choice of which S-graph edges are active (not stopped).

The sAMY hub (in-degree 19, out-degree 22 in Q_{75}) is the unique vertex through which all elements of Λ_{red} pass, making it the central vertex of $\mathbb{S}$.

Definition 11.8 ([24, §3])**.** A *perverse schober* $\mathcal{F}$ parametrised by ribbon graph $\mathbf{G}$ is a functor $\mathcal{F} \colon \text{Exit}(\mathbf{G}) \to \text{St}$ from the exit-path category to the ∞-category St of stable ∞-categories, subject to local exactness conditions. Its *stable ∞-category of global sections* is $\Gamma(\mathbf{G}, \mathcal{F}) = \lim(\text{Exit}(\mathbf{G}) \xrightarrow{\mathcal{F}} \text{St})$.

Proposition 11.9. *The global sections category $\Gamma(\mathbb{S}, \mathcal{F})$ of the perverse schober parametrised by $\mathbb{S}$ of Construction 11.7 satisfies $\Gamma(\mathbb{S}, \mathcal{F}) \simeq \mathcal{W}(\Sigma_Q, \Lambda^+ \cup \Lambda^-)$ (Sector A). More generally, $\mathcal{W}(\Sigma_Q, \Lambda_S) \simeq \Gamma(\mathbb{S}_S, \mathcal{F}_S)$ for each $S \in \{A, B, C, D\}$.*

Theorem 11.10 (Flips are GPS restriction maps)**.** *The GPS restriction maps are transported schobers under S-graph flips:*

$$\rho_{AB} \colon \mathcal{W}(\Sigma_Q, \Lambda^+) \longrightarrow \mathcal{W}(\Sigma_Q, \Lambda^+ \cup \Lambda^-) \qquad \text{(backward flip of } \Lambda^- \text{ edges)},$$
$$\rho_{AC} \colon \mathcal{W}(\Sigma_Q, \Lambda^-) \longrightarrow \mathcal{W}(\Sigma_Q, \Lambda^+ \cup \Lambda^-) \qquad \text{(forward flip of } \Lambda^+ \text{ edges)}.$$

The transported schober $\mathcal{F}^\flat$ inherits an arc system kit ([24, §4.2]), and $\Gamma(\mathbb{S}, \mathcal{F}) \simeq \Gamma(\mathbb{S}^\flat, \mathcal{F}^\flat)$. The spectral inertia $\rho(\mathcal{W}_A) = \rho(\mathcal{W}_B) = 2^{1/3}$ is the statement that flipping Λ^- is a trivial tilt*: the Λ^- edges generate no new simples in the tilted heart.*

Remark 11.11 (Arc system kit = AU context extension). The arc system kit of [24, §4.2] is a recipe assigning local sections of $\mathcal{F}$ to arc segments such that compatible sections glue to global sections. An AU context extension $T_\alpha \subset T_{\alpha\beta}$ is an arc system kit in this sense, with the restriction map $\rho_{\alpha\beta} \colon \mathcal{W}(T_{\alpha\beta}) \to \mathcal{W}(T_\alpha)$ as the gluing condition. The AU guarantees compatible local sections assemble to global sections without constructing the global object.

11.4 Renkin–Crone weights as central charges of stability conditions

Theorem 11.12 ([24, Theorem 1.1]). *Let $\mathcal{C}(\mathbb{S},\mathcal{F})$ be the stable ∞-category generated by $\Gamma_{\mathbb{S}}$. There is a biholomorphism*

$$\mathrm{FQuad}^S(\mathbf{S}_w) \xrightarrow{\sim} \mathrm{Stab}\big(\mathcal{C}(\mathbb{S},\mathcal{F})\big)$$

from framed quadratic differentials to Bridgeland stability conditions, onto a union of connected components.

Corollary 11.13 (Renkin–Crone weights as central charges). *The Renkin–Crone flow weight c_{XYZ} of a relation $f_{XY} \cdot f_{YZ} = c_{XYZ} \cdot f_{XZ}$ is the modulus $|Z(\gamma_{XYZ})|$ of the central charge of γ_{XYZ} in $\sigma = (Z,\mathcal{P}) \in \mathrm{Stab}(\mathcal{C}(\mathbb{S},\mathcal{F}))$. The GPS spectral radii are the Perron–Frobenius eigenvalues of $(|Z(\gamma)|)_{\gamma\in\Gamma_{\mathbb{S}_S}}$.*

Theorem 11.14 (Wall-crossing = phase transition = crisis boundary). *The spectral jump $\rho(\mathcal{W}_A) = \rho(\mathcal{W}_B) = 2^{1/3} \to \rho(\mathcal{W}_C) = 1.909$ is a wall-crossing in $\mathrm{Stab}(\mathcal{C}(\mathbb{S},\mathcal{F}))$: Sectors A and C lie in different connected components. Since $H^2(\mathrm{Cone}(\rho_{AC})) \neq 0$ (Theorem 6.3, third case), the $A \to C$ transition is irreversible at the A_∞-enhanced level (Definition 11.4).*

Theorem 11.15 (GPS sectors detect filtration extension structure). *The four GPS sectors form a filtration $\mathcal{W}_A \subset \mathcal{W}_B \subset \mathcal{W}_C \subset \mathcal{W}_D$ of the fully-open category $\mathcal{W}(\Sigma_Q,\varnothing)$, with restriction maps as the inclusion functors. The spectral radii $\rho(B_{\Lambda_S})$ are* not *Postnikov k-invariants in the strict sense (which would require cohomology classes controlling extensions); rather, each spectral radius* detects the categorical extension class geometry *of the corresponding filtration layer: the growth rate of the Grothendieck group $K_0(\mathcal{W}_S)$ under the shift functor encodes how the heart $\mathcal{H}_{\Lambda_S}$ extends across the layer.*

$$\begin{tikzcd}
& \mathcal{W}_A \arrow[dl, "\rho_{AB}"'] \arrow[dr, "\rho_{AC}"] \arrow[dd, "\rho_{AD}"] & \\
\mathcal{W}_B \arrow[dr, "\rho_{BD}"'] & & \mathcal{W}_C \arrow[dl, "\rho_{CD}"] \\
& \mathcal{W}_D &
\end{tikzcd}$$

$\rho(B_S)$ detects extension class of Layer S.

Theorem 11.16 (Golden ratio as minimal stability condition). *The minimal S-graph $\mathbb{S}_D = \{\mathrm{LA} \leftrightarrow \mathrm{sAMY}\}$ has a unique framed quadratic differential with two simple poles whose central charges satisfy $|Z(\gamma_1)|^2 = |Z(\gamma_1)| + |Z(\gamma_2)|$, forcing $|Z(\gamma_i)| = \varphi$ by $\lambda^2 = \lambda + 1$. The category $\mathcal{C}(\mathbb{S}_D,\mathcal{F})$ is* 2-Calabi–Yau*: two simples with one non-trivial extension form an A_2 Auslander–Reiten quiver with Coxeter number* 3 *and growth rate φ. This is consistent with the six-decimal confirmation $\rho(\mathcal{W}_D) = 1.618034$ of [1] (Prediction P2) and Conjecture 9.2.*

11.5 Complex of Fukaya categories: Dyckerhoff's categorified Dold–Kan

Theorem 11.17 ([25, Main Theorem]). *The categorified normalised chains functor $\mathcal{C}$ furnishes an equivalence*

$$\mathcal{C}\colon \mathrm{St}_\Delta \overset{\simeq}{\longleftrightarrow} \mathbf{Ch}_{\geq 0}(\mathrm{St})\colon \mathcal{N}$$

between 2-simplicial stable ∞-categories and connective chain complexes of stable ∞-categories. This is one direction of the identification in Theorem 11.1(3): the complex realization $\mathcal{W}_\bullet$ and the global-sections realization $\mathcal{C}(\mathbb{S}_\Lambda, \mathcal{F})$ are identified by $\mathcal{C}$.

Remark 11.18 (GPS triangle as 2-simplex in $\mathcal{N}$). A 2-simplex in $\mathcal{N}(\mathcal{B}_\bullet)$ has objects $X_{01}, X_{02}, X_{12} \in \mathcal{B}_1$, $X_{012} \in \mathcal{B}_2$, with $d(X_{012}) = \mathrm{tot}(X_{01} \to X_{02} \to X_{12})$. The GPS triangle $A \to B \to D$ is this 2-simplex with $X_{01} = \rho_{AB}$, $X_{12} = \rho_{BD}$, $X_{02} = \rho_{AD}$, and $d(X_{012}) = \mathrm{Cone}(\rho_{AB})$ ([25, §1, Example (II)]).

Remark 11.19 (GPS descent as Waldhausen's $S_\bullet$-construction). By [25, §1, Example (II)], $\mathcal{N}(\mathcal{B}[1])$ is the ∞-categorical Waldhausen $S_\bullet$-construction. The GPS descent sequence $\mathcal{W}(\Sigma_Q, \Lambda \cup \{e\}) \to \mathcal{W}(\Sigma_Q, \Lambda) \to \mathrm{Cone}(e) \to \mathcal{W}(\Sigma_Q, \Lambda \cup \{e\})[1]$ is Waldhausen's sequence for the restriction functor of stop removal.

Proposition 11.20 (Conservativity and crisis detection). *By [25, Proposition 2.5], $\mathcal{C}$ is conservative: a morphism is an equivalence iff $\mathcal{C}$ maps it to an equivalence. Applied to $\rho_{\alpha\beta}$ at the A_∞-enhanced level:*

$$\rho_{\alpha\beta} \text{ is a quasi-equivalence} \iff H^2\big(\mathrm{Cone}(\rho_{\alpha\beta})\big) = 0 \iff \text{the transition is reversible.}$$

The non-vanishing $H^2(\mathrm{Cone}(\rho_{AC})) \neq 0$ confirmed in [1] is the DK-conservativity obstruction at the A_∞-enhanced level, consistent with the clinical irreversibility of the $A \to C$ transition. Note: this criterion requires the A_∞-enhanced category (Definition 11.4), not merely $D^b(\mathcal{W})$ where $H^2(\mathrm{Cone}(\rho))$ could vanish spuriously.

Remark 11.21 (Higher syzygies as K-theory classes). By [25, Formula (1.1)], $\Omega^k|\mathcal{N}(\mathcal{B}[k])^{\simeq}| \simeq K(\mathcal{B})$. The operations m_k for $k \geq 3$ are syzygies in K_k of the Fukaya complex; the $\approx$4000 relations $f_{XY} \cdot f_{YZ} = c \cdot f_{XZ}$ are the m_2 part. These syzygies are *invisible* in the triangulated category $D^b(\mathcal{W})$ but *visible* in $\mathbf{Der}_{2,1}(T_\Lambda)$ (Definition 11.4), consistent with Remark 11.3.

11.6 The complete correspondence

Remark 11.22 (Why CHQ and Dyckerhoff do not replace the AU). Theorems 11.23 and 11.28 provide mathematical structures; they do not prescribe when to compute global sections or which flip to apply. The AU provides three additional ingredients:

(i) **Logical substrate.** The inert quiver data (Q_0, Q_1, R) exists before any stop choice, held as a finitely-presented free structure [14].
(ii) **On-demand crystallisation.** A stop choice Λ triggers the CHQ construction of Level 3 and the A_∞-module construction of Level 4 without pre-computing either. Nothing exists until queried.

Table 1: Dictionary: Christ–Haiden–Qiu [24], Dyckerhoff [25], and the AU framework of this paper. Level column refers to Theorem 11.1.

CHQ [24]	**Dyckerhoff [25]**	**AU (this paper)**	**Biological meaning**	**Level**
S-graph $\mathbb{S}_\Lambda$	—	Skeleton $\mathbb{S}_\Lambda$	Stop architecture	1
Simple-minded coll. $\Gamma_{\mathbb{S}}$	Degree-0 of $\mathcal{W}_\bullet$	Heart $\mathcal{H}_\Lambda$	Baseline phase	2
$\mathcal{C}(\mathbb{S}_\Lambda, \mathcal{F})$	$\mathcal{W}_\bullet$ (complex realization)	$\mathcal{W}(\Sigma_Q, \Lambda)$	GPS sector category	3 (parallel)
S-graph flip	Differential $d_k = m_k$	GPS restriction map ρ	Drug effect onset	3
Arc system kit	DK nerve $\mathcal{N}$	AU context extension	Effect-driven context	3
$\mathrm{Stab}(\mathcal{C}(\mathbb{S}, \mathcal{F}))$	K-theory space $\lvert\mathcal{N}(\mathcal{B}[k])^{\simeq}\rvert$	$\rho(B_\Lambda)$	Phase invariant	3
Wall-crossing	$H^2(\mathrm{Cone}(\rho)) \neq 0$ at enh. level	Crisis boundary	Irreversible transition	4
$\mathrm{FQuad}^S \xrightarrow{\sim} \mathrm{Stab}$	$\Omega^k\lvert\mathcal{N}(\mathcal{B}[k])^{\simeq}\rvert \simeq K$	Renkin–Crone $\to \rho$	Data $\to$ prediction	3–4
—	m_k, $k \geq 3$, syzygies	$\mathbf{Der}_{2,1}(T_\Lambda)$, k-invariants	Higher obstruction	4
$\lambda^2 = \lambda + 1$ on $\mathbb{S}_D$	—	$\rho(\mathcal{W}_D) = \varphi$, 2-CY	Universal golden ratio	2–3

(iii) **Consistency guarantee.** The AU's internal logic ensures any two on-demand computations sharing a context extension are related by a restriction map in $\mathbf{Der}_{2,1}$, without constructing the global colimit (Theorem 4.6).

The AU is the logical type system over the CHQ–Dyckerhoff mathematical structure that makes scaling to $N = 27$ and beyond feasible (Theorem 4.3).

11.7 Renkin–Crone weights as central charges of stability conditions

Recall that a Bridgeland stability condition [23] on a triangulated category $\mathcal{D}$ is a pair $\sigma = (Z, \mathcal{P})$ of a central charge $Z\colon K_0(\mathcal{D}) \to \mathbb{C}$ and a full slicing $\mathcal{P}$. The space $\mathrm{Stab}(\mathcal{D})$ is a complex manifold.

Theorem 11.23 ([24, Theorem 1.1]). *Let $\mathcal{C}(\mathbb{S},\mathcal{F})$ be the stable ∞-category generated by the simple-minded collection $\Gamma_{\mathbb{S}}$ associated with the S-graph edges. There is a biholomorphism*

$$\mathrm{FQuad}^S(\mathbf{S}_w) \xrightarrow{\sim} \mathrm{Stab}\big(\mathcal{C}(\mathbb{S},\mathcal{F})\big)$$

from the space of framed quadratic differentials on $\mathbf{S}_w$ to the space of Bridgeland stability conditions on $\mathcal{C}(\mathbb{S},\mathcal{F})$, onto a union of connected components.

Corollary 11.24 (Renkin–Crone weights as central charges). *Under Theorem 11.23, the Renkin–Crone flow weight c_{XYZ} of a path-composition relation $f_{XY}\cdot f_{YZ} = c_{XYZ}\cdot f_{XZ}$ is the modulus $|Z(\gamma_{XYZ})|$ of the central charge of the corresponding object γ_{XYZ} in the stability condition $\sigma = (Z,\mathcal{P}) \in \mathrm{Stab}(\mathcal{C}(\mathbb{S},\mathcal{F}))$. The GPS spectral radii are the Perron–Frobenius eigenvalues of the central charge vector restricted to the simple-minded collection $\Gamma_{\mathbb{S}_S}$:*

$$\rho(B_{\Lambda_S}) \;=\; \text{PF-eigenvalue of } \big(|Z(\gamma)|\big)_{\gamma\in\Gamma_{\mathbb{S}_S}}.$$

Corollary 11.25 (GPS sectors as stability phase diagram). *Each GPS sector $S \in \{A,B,C,D\}$ is a connected component $\mathrm{Stab}_S \subset \mathrm{Stab}(\mathcal{C}(\mathbb{S}_S,\mathcal{F}_S))$. The GPS restriction maps act by wall-crossing: $\rho^*_{AB}\sigma \in \mathrm{Stab}_B$, $\rho^*_{AC}\sigma \in \mathrm{Stab}_C$, $\rho^*_{AD}\sigma \in \mathrm{Stab}_D$.*

Theorem 11.26 (Wall-crossing = phase transition = crisis boundary). *The spectral jump $\rho(\mathcal{W}_A) = \rho(\mathcal{W}_B) = 2^{1/3} \;\longrightarrow\; \rho(\mathcal{W}_C) = 1.909$ is a wall-crossing in $\mathrm{Stab}(\mathcal{C}(\mathbb{S},\mathcal{F}))$: sectors A and C lie in different connected components, separated by the locus $\mathrm{Im}(Z(\gamma)) = 0$ for a stable object $\gamma \in \Gamma_{\mathbb{S}}$. Since $H^2(\mathrm{Cone}(\rho_{AC})) \neq 0$ (Theorem 6.3, third case), the $A \to C$ transition is irreversible: this is the categorical signature of the opioid crisis boundary.*

Theorem 11.27 (Golden ratio as minimal stability condition). *The minimal S-graph $\mathbb{S}_D = \{\mathrm{LA} \leftrightarrow \mathrm{sAMY}\}$ has a unique framed quadratic differential with two simple poles. Its central charges satisfy $|Z(\gamma_1)| = |Z(\gamma_2)|$ and $|Z(\gamma_1)|^2 = |Z(\gamma_1)| + |Z(\gamma_2)|$, forcing $|Z(\gamma_i)| = \varphi$ by the characteristic equation $\lambda^2 = \lambda + 1$. Equivalently, non-backtracking walks on the single-edge ribbon graph satisfy the Fibonacci recursion, and φ is the Perron–Frobenius eigenvalue of the Hashimoto matrix. This is consistent with the six-decimal-place confirmation $\rho(\mathcal{W}_D) =$ 1.618034 of [1] (Prediction P2) and with the universal golden ratio Conjecture 9.2.*

11.8 Complex of Fukaya categories: Dyckerhoff's categorified Dold–Kan

The local Fukaya complex $\mathcal{W}^{\alpha}_{\bullet}$ of §3 is not merely a notational device; it is an instance of Dyckerhoff's categorified Dold–Kan correspondence [25].

Theorem 11.28 ([25, Main Theorem]). *The categorified normalised chains functor $\mathcal{C}$ furnishes an equivalence*

$$\mathcal{C}\colon \mathrm{St}_\Delta \overset{\simeq}{\longleftrightarrow} \mathbf{Ch}_{\geq 0}(\mathrm{St})\colon \mathcal{N}$$

between 2-simplicial stable ∞-categories and connective chain complexes of stable ∞-categories, with explicit inverse the categorified Dold–Kan nerve $\mathcal{N}$.

Remark 11.29 (GPS triangle as 2-simplex in $\mathcal{N}$). A 2-simplex in the DK nerve $\mathcal{N}(\mathcal{B}_\bullet)$ consists of objects $X_{01}, X_{02}, X_{12} \in \mathcal{B}_1$ and $X_{012} \in \mathcal{B}_2$ with $d(X_{012}) = \mathrm{tot}(X_{01} \to X_{02} \to X_{12})$. The GPS sector triangle $A \to B \to D$ is this 2-simplex:

$$X_{01} = \rho_{AB}, \quad X_{12} = \rho_{BD}, \quad X_{02} = \rho_{AD},$$

and the totalization $d(X_{012}) = \mathrm{Cone}(\rho_{AB})$ is the mapping cone of the $A \to B$ restriction map (cf. [25, §1, Example (II)]).

Remark 11.30 (GPS descent as Waldhausen's $S_\bullet$-construction). By [25, §1, Example (II)], $\mathcal{N}(\mathcal{B}[1])$ is the ∞-categorical Waldhausen $S_\bullet$-construction for an exact functor $f\colon \mathcal{B}_1 \to \mathcal{B}_0$. The GPS descent sequence

$$\mathcal{W}(\Sigma_Q, \Lambda \cup \{e\}) \longrightarrow \mathcal{W}(\Sigma_Q, \Lambda) \longrightarrow \mathrm{Cone}(e) \longrightarrow \mathcal{W}(\Sigma_Q, \Lambda \cup \{e\})[1]$$

for stop insertion at edge e is Waldhausen's sequence applied to the restriction functor of stop removal.

Proposition 11.31 (Conservativity and crisis detection). *By [25, Proposition 2.5], the normalised chains functor $\mathcal{C}$ is* conservative*: a morphism of* 2*-simplicial stable ∞-categories is an equivalence if and only if $\mathcal{C}$ maps it to an equivalence of chain complexes. Applied to the GPS restriction map $\rho_{\alpha\beta}$:*

$$\rho_{\alpha\beta} \text{ is a quasi-isomorphism} \iff H^2\big(\mathrm{Cone}(\rho_{\alpha\beta})\big) = 0 \iff \text{the transition is reversible.}$$

This is the mathematical foundation of the crisis detection in Theorem 6.3: the non-vanishing $H^2(\mathrm{Cone}(\rho_{AC})) \neq 0$ confirmed in [1] is the DK-conservativity obstruction to the $A \to C$ transition being a quasi-isomorphism, consistent with its clinical irreversibility.

Remark 11.32 (Higher A_∞ syzygies as K-theory classes). Formula (1.1) of [25] states $\Omega^k|\mathcal{N}(\mathcal{B}[k])^{\simeq}| \simeq K(\mathcal{B})$. The higher A_∞ operations m_k for $k \geq 3$ are syzygies appearing as classes in K_k of the Fukaya complex; the $\approx$4000 quiver relations $f_{XY} \cdot f_{YZ} = c \cdot f_{XZ}$ are the degree-2 part (m_2). The GPS sectors are the Postnikov sections of the resulting K-theory tower, with spectral radii as the Postnikov k-invariants:

$$\begin{tikzcd}
& \mathcal{W}_A \arrow[dl, "\rho_{AB}"'] \arrow[dr, "\rho_{AC}"] \arrow[dd, "\rho_{AD}"] & \\
\mathcal{W}_B \arrow[dr, "\rho_{BD}"'] & & \mathcal{W}_C \arrow[dl, "\rho_{CD}"] \\
& \mathcal{W}_D &
\end{tikzcd} \qquad \rho(B_S) = \text{PF-eigenvalue of } K_0(\mathcal{W}_S).$$

11.9 The complete correspondence

Table 2 gives the precise identification across the three frameworks.

Table 2: Dictionary: Christ–Haiden–Qiu [24], Dyckerhoff [25], and the AU framework of this paper.

CHQ [24]	**Dyckerhoff [25]**	**AU (this paper)**	**Biological meaning**
S-graph edges	Degree-0 complex	GPS Sector A ($\Lambda^+ \cup \Lambda^-$)	Baseline brain state
S-graph flip	Normalised chains $\mathcal{C}$	GPS restriction map ρ_{AB}	Drug effect onset
Arc system kit	DK nerve $\mathcal{N}$	AU context extension	Effect-driven context
Simple-minded coll. $\Gamma_{\mathbb{S}}$	Waldhausen $S_\bullet$	GPS sector A/B/C/D	Categorical phase
Perverse schober $\mathcal{F}$	Chain complex of St-cats	$\mathcal{W}^\alpha_\bullet(\Sigma_Q, \Lambda_\bullet)$	Multi-homology AU object
Transported schober $\mathcal{F}^\flat$	$\mathcal{C} \circ \mathcal{N} \simeq \mathrm{id}$	Emergent $\mathcal{W}(\Sigma_Q, \Lambda')$	Post-intervention state
$\mathrm{Stab}(\mathcal{C}(\mathbb{S}, \mathcal{F}))$	K-theory space	$\rho(B_\Lambda)$	Phase invariant
$\mathrm{FQuad}^S \xrightarrow{\sim} \mathrm{Stab}$	$\Omega^k \lvert\mathcal{N}(\mathcal{B}[k])^\simeq\rvert \simeq K$	Renkin–Crone $\to \rho$	Data $\to$ prediction
Wall-crossing	$H^2(\mathrm{Cone}(\rho)) \neq 0$	Crisis boundary	Irreversible transition
$\lambda^2 = \lambda+1$ on $\mathbb{S}_D$	Eilenberg–MacLane $K(\mathbb{Z}, 1)$	$\rho(\mathcal{W}_D) = \varphi$	Universal golden ratio

Remark 11.33 (Why CHQ and Dyckerhoff do not replace the AU). Theorems 11.23 and 11.28 are mathematical structures; they do not prescribe *when* to compute global sections or *which* flip to apply. The AU provides three additional ingredients absent from both papers:

(i) **Logical substrate.** The inert quiver data (Q_0, Q_1, R) exists before any stop choice is made, held as a finitely-presented free structure [14].

(ii) **On-demand crystallisation.** A stop choice Λ triggers the CHQ construction of $\mathcal{C}(\mathbb{S}_\Lambda, \mathcal{F}_\Lambda)$ and the DK nerve $\mathcal{N}(\mathcal{W}^\alpha_\bullet(\Lambda))$ without pre-computing either. Nothing exists until queried.

(iii) **Consistency guarantee.** The AU's internal logic ensures any two on-demand computations sharing a context extension are related by a restriction map in $\mathbf{Der}_{2,1}$, without constructing the global colimit. This is the content of Theorem 4.6.

The AU is the logical type system over the CHQ–Dyckerhoff mathematical structure that makes scaling to $N = 27$ (and beyond) feasible (Theorem 4.3).

12 The A_∞-functor lifting problem

The central open question connecting [1] to the present paper concerns whether the GPS restriction maps lift from H^0-functors to full A_∞-functors. By Theorem 6.3, this question is equivalent to asking which row of the additive classification table applies to each region pair — and the AU is the framework that makes this question well-posed, since it guarantees the derived $(2,1)$-stack $\mathbf{Der}_{2,1}(T_\alpha)$ is well-defined and that i^* exists for every context extension.

Definition 12.1. The restriction map $\rho_{\alpha\beta} : \mathcal{W}^\beta_\bullet \to \mathcal{W}^\alpha_\bullet$ is an H^0*-functor* if it is well-defined at the level of cohomology categories. It is a *full* A_∞*-functor* if it commutes with all m_k operations up to coherent homotopy.

Remark 12.2. The ρ_{AC} map of [1] is confirmed as an H^0-functor by the GPS transfer theorem. Whether it lifts to a full A_∞-functor depends on whether the 62–102 defect triples form Čech 2-cocycles in $\check{H}^2(\mathcal{U}, \mathcal{H}om_{A_\infty})$ obstructing the pushout. By Theorem 6.3, this determines whether ρ_{AC} falls in the second row (partial addition, H^0 only) or, if the cocycles are coboundaries, in the first row (full addition, quasi-isomorphism). The AU internal to which both $\mathcal{W}^A_\bullet$ and $\mathcal{W}^C_\bullet$ live is the context AU$\langle T_{AC}\rangle$ generated by the Sector A and Sector C stop configurations; the pullback $i^* : \mathbf{Der}_{2,1}(T_{AC}) \to \mathbf{Der}_{2,1}(T_A)$ is what makes the comparison well-defined.

The biological consequences now follow directly from Theorem 6.3:

- *Defect triples are coboundaries:* $\rho_{\alpha\beta}$ is a full A_∞-functor (first row). The AU projection maps preserve the complete A_∞ structure; recovery restores both observable behaviour (H^0) and all higher-order pathway correlations ($m_3, \ldots, m_6$).
- *Defect triples are genuine cocycles, $\rho_{\alpha\beta}$ is H^0:* second row. Recovery restores observable behaviour but higher-order correlations are permanently deformed. This is a categorical signature of persistent pharmacodynamic change.
- *Defect triples obstruct even H^0:* third row. No transfer is possible; regions are categorically independent (Remark 6.4).

The 27-region pipeline is designed to decide between these cases: with enough region pairs to compute defect triples, the cohomology class [defect] $\in \check{H}^2$ becomes computable and the row of Theorem 6.3 that applies to each region pair is determined.

Theorem 12.3 (Postnikov proxy for $H^2(\mathrm{Cone}(\rho))$)**.** *For the $N = 7$ BALBc connectome, the categorical obstruction $H^2(\mathrm{Cone}(\rho_{\alpha\beta})) \neq 0$ is detected by the growth rate of newly active walks in the Postnikov tower. Define the* Postnikov differential

$$\Delta_k(\alpha \to \beta) \;=\; |\mathcal{W}_k(\beta)| - |\mathcal{W}_k(\alpha)|$$

where $|\mathcal{W}_k(S)|$ is the number of non-backtracking walks of length k with active first edge in sector S. The computed values for $k = 1, \ldots, 6$ are:

Transition	Δ_1	Δ_2	Δ_3	Δ_4	Δ_5	Δ_6	*Signal*
$A \to B$	*4*	*8*	*10*	*18*	*58*	*86*	*inertia* ($H^2 = 0$)
$A \to C$	*4*	*11*	*12*	*29*	*66*	*103*	*obstruction* ($H^2 \neq 0$)
$A \to D$	*6*	*14*	*16*	*34*	*92*	*138*	*obstruction* ($H^2 \neq 0$)
$B \to D$	*2*	*6*	*6*	*16*	*34*	*52*	*obstruction* ($H^2 \neq 0$)
$C \to D$	*2*	*3*	*4*	*5*	*26*	*35*	*partial*

The transition $A \to B$ *(spectral inertia,* $\rho(\mathcal{W}_A) = \rho(\mathcal{W}_B)$*) has* $\Delta_1 = \Delta_1(A \to C) = 4$*: the two transitions are* indistinguishable at Postnikov level $k = 1$. *The crisis wall* $A \to C$ *first becomes visible at* $k = 2$*, where* $\Delta_2(A \to C) = 11 > 8 = \Delta_2(A \to B)$.

Remark 12.4 (Two walk space constructions)*.* Two distinct walk space constructions appear in the computational framework, with different chain complex properties:

(i) *Active-first-edge walks* (`cone_h2.jl`): $\mathcal{W}_k(S)$ = walks whose first edge is not stop-blocked. This is *not* a chain complex: the Waldhausen boundary $d\colon \mathcal{W}_2(S) \to \mathcal{W}_1(S)$ maps some walks to walks starting with stopped edges, which are not in $\mathcal{W}_1$. Concretely, 12 of 25 length-2 walks in Sector A have a stopped second edge, so $d^2 \neq 0$. This construction computes the Postnikov differential $\Delta_k(\alpha \to \beta)$ of Theorem 12.3.

(ii) *Full-adjacency walks* (`postnikov_tower.jl`): $\mathcal{W}_k(S)$ = walks whose first edge is active, but subsequent edges may be stopped (they appear as column sources in B_{Λ_S}, matching the Hashimoto semantics). This *is* a chain complex: $D^2 = 0$ verified for $k = 1, \ldots, 6$ in all four GPS sectors. The Waldhausen $S_\bullet$-construction differential cancels correctly because the full-adjacency walks form a closed module under the tail-minus-head differential.

The GPS sectors form a Postnikov filtration $\mathcal{W}_A \subset \mathcal{W}_B \subset \mathcal{W}_C \subset \mathcal{W}_D$ as stable ∞-categories and as Waldhausen complexes (construction (ii)), but not as active-first-edge complexes (construction (i)).

Remark 12.5 (Four computational detections of $H^2(\mathrm{Cone}(\rho))$)*.* The categorical obstruction $H^2(\mathrm{Cone}(\rho_{\alpha\beta}))$ of Theorem 6.3 is detected computationally by four independent criteria, all confirmed for the $N = 7$ BALBc quiver:

(a) **Spectral inertia** $\rho(\mathcal{W}_A) = \rho(\mathcal{W}_B)$: $H^2 = 0$ for $A \to B$; no obstruction. Confirmed: $|\rho(\mathcal{W}_B) - \rho(\mathcal{W}_A)| = 0.000000$.

(b) **Spectral jump** $\rho(\mathcal{W}_A) \neq \rho(\mathcal{W}_C)$: $H^2 \neq 0$ for $A \to C$; crisis obstruction. Confirmed: $\rho(\mathcal{W}_C)/\rho(\mathcal{W}_A) = 1.5152 > 1.2$.

(c) **Boundary rank** $\mathrm{rank}(B_C - B_A) = |\Lambda^+| = 6$: newly opened morphisms equal the stop set size. Confirmed: newly_opened $= 6 = |\Lambda^+|$.

(d) **Postnikov differential** $\Delta_2(A \to C) > \Delta_2(A \to B)$: the crisis wall is invisible at $k = 1$ but detectable at $k = 2$. Confirmed: $11 > 8$.

Criteria (a)–(c) are the rigorous content of Theorem 6.3. Criterion (d) is a new structural prediction of the Postnikov tower (Theorem 12.3): the crisis becomes visible precisely at the second Postnikov level, consistent with the $\mathrm{Gr}(2, 4)$ Schubert cell structure (Theorem 8.7).

13 Open questions

Q1. **A_∞-functor lifting.** Do the GPS restriction maps $\rho_{\alpha\beta}$ lift from H^0-functors to full A_∞-functors? Equivalently: are the defect triples Čech 2-cocycles or coboundaries in $\check{H}^2(\mathcal{U}, \mathcal{H}om_{A_\infty})$?

Q2. **Universal golden ratio.** Does Conjecture 9.2 hold for any connected region pairs in the 27-region Allen Brain Atlas connectome beyond LA–sAMY? The test procedure in Remark 9.3 is computationally straightforward: compute $\rho(\mathcal{W}^{\min}_{(\alpha,\beta)})$ for all 351 region pairs, check the Fibonacci condition, and record which pairs satisfy $|\rho - \varphi| < 10^{-5}$. A null result (no additional pairs) confirms LA–sAMY as topologically unique in the network.

Q3. **Bockstein computation.** Recall from [1] that the LA$\leftrightarrow$sAMY edge weight $w_{\mathrm{LA,sAMY}} = 9752/100$ satisfies $v_5(w) = -2$, placing it in $\mathbb{Z}_5[1/5] \setminus \mathbb{Z}_5$ and obstructing integrality over $\mathbb{Z}_5$. The open question is the explicit computation of the Bockstein differential $\beta([b_0]) \in H^2(L_{\mathrm{LA}}; \mathbb{F}_p)$ for $p = 5$, to determine whether the 5-adic obstruction at the LA–sAMY edge is a genuine Čech cocycle (permanent structural obstruction) or a coboundary (removable by a change of bounding cochain).

Q4. **AU tensor product.** After clearing all p-adic poles by lifting to $\mathbb{Z}[1/10]$, does the tensor product $\mathcal{W}^\alpha_\bullet \otimes_{\mathbb{Z}[1/10]} \mathcal{W}^\beta_\bullet$ exist as a curved A_∞-category? Remark 3.12 shows that the representation construction $\mathrm{Rep}(A_3, \mathcal{W}^\alpha_\bullet)$ is the primary object available over $\mathbb{Z}_5$; the tensor construction is its derived enhancement over $\mathbb{Z}[1/10]$. The open question is whether the two constructions agree on $\mathbb{Z}[1/10] \cap \mathbb{Z}_5 = \mathbb{Z}[1/10]_5$.

Q5. **27-region spectral ordering.** The $N = 7$ spectral ordering $\rho(A) = \rho(B) < \rho(D) < \rho(C)$ is non-monotone in stop count. Does an analogous non-monotone ordering hold for the $N = 27$ network, and if so, which stop configurations are spectrally inert?

Q6. **Narcan rank-4 prediction.** The AU framework predicts that opioid-reversal by Naloxone has $\mathrm{rank}(B_{T'_{\mathrm{baseline}}} - B_{T_{\mathrm{opioid}}}) = 4$. Test this against PAG-region connectome data in the 27-region network.

Q7. **Incommensurable contexts prediction.** By Theorem 6.3 and Remark 6.4, region pairs (α, β) for which the defect triple class $[\mathrm{defect}] \in \check{H}^2(\mathcal{U}, \mathcal{H}om_{A_\infty})$ is nonzero are categorically independent: no additive interaction of A_∞ pathway structure is possible. *Testable prediction.* Compute [defect] for all 351 region pairs in the 27-region Allen Atlas network. For pairs with nonzero class, the drug effects in those regions should show no A_∞-level superposition. Test against existing drug interaction databases by checking whether pharmacologically independent drug pairs (opioid + SSRI, or similar) correspond to region pairs with nonzero defect class. A positive correlation would be strong evidence for the AU framework's predictive power.

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